# Weekly Research Candidate - 2026-07-30

Candidate - awaiting-eval - not stable guidance

## Status and scope

> **CANDIDATE - not stable guidance, no authority.** Not applied to Runtime/stable, Core, bootstraps, global pointers, sibling repos, the source registry, or the model catalog. Ends at `awaiting-eval`; promotion is separate and human-approved.

Second weekly run, building on 2026-07-20: all 84 prior open questions were routed into the buckets and last week's discovered sources were pre-loaded. Multi-modal this week (academic deep-dive, YouTube, Reddit) plus the now-due model-catalog refresh (catalog expired 2026-07-26). Result: **254 atomic claims** (250 accepted, 4 rejected), **73 source-registry changes**, **33 model-catalog facts**, 82 drift notes, 74 open questions.

## Security notice (read first)

> The auto-mode monitor flagged the **claude-hooks-permissions** research and verify agents as possible reconnaissance into the permission classifier's internals. The bucket documented official `code.claude.com` permission mechanics for the contract's Permissions section - assessed as a **likely false positive** (defensive documentation from official sources). Its **24 claims are tagged** `security_review_required` and are gated behind blocking eval case **EV-SEC**; they must be human-cleared before informing any promoted text.

## Method

11 buckets, each research -> independent verify. Rules enforced: version-sensitive claim needs a current tier-1 primary source (else `unsupported`); model-id/number needs an official provider source; a citation that does not directly support the statement is `unsupported`; tier 4-5 (incl. YouTube/Reddit) is a LEAD, corroborated to tier 1-2 or filed as an open question. The orchestration hard-fails on duplicate ids, missing verdicts, or an empty bucket. 22 agents, 0 errors.

## Claim coverage

| Verdict | Count |
|---|---:|
| confirmed | 244 |
| corrected | 6 |
| unsupported | 4 |

Accepted confidence: **high 243**, medium 7. Version-sensitive accepted: **244**.

| Bucket | Accepted | Rejected |
| --- | ---: | ---: |
| claude-loops-scheduling | 25 | 0 |
| claude-hooks-permissions (SEC-review) | 24 | 0 |
| claude-context-memory-skills | 30 | 0 |
| codex-config-hooks | 23 | 0 |
| codex-exec-sandbox | 24 | 0 |
| security-sandboxing-injection | 9 | 0 |
| evals-judging | 14 | 4 |
| academic-deep-dive | 30 | 0 |
| youtube-lane | 13 | 0 |
| reddit-lane | 25 | 0 |
| model-catalog-refresh | 33 | 0 |

## Model-catalog refresh (candidate providers patch)

The live catalog expired 2026-07-26; `models-providers.patch.json` captures the verified facts (candidate only - EV-MODEL re-verifies each before any catalog write). Headline changes:

- Claude flagship is now Opus 5 (claude-opus-5); Opus 4.8 (claude-opus-4-8) is now LEGACY; Opus 4.1 deprecated, retires 2026-08-05.
- Claude effort levels: low|medium|high|xhigh|max (high=default). 'ultracode' is a Claude Code SESSION setting (sends xhigh + orchestrates dynamic workflows), NOT an API effort level.
- OpenAI/Codex: GPT-5.6 Sol/Terra/Luna (1.05M ctx, 128K out); Codex CLI 0.146.0 (2026-07-29). platform.openai.com/docs/models now 301-redirects to developers.openai.com/api/docs/models.

Model-catalog claims:

- **`model-catalog-refresh-01`** - Claude Opus 5 is now the recommended current flagship for complex agentic coding: Claude API id and alias are both `claude-opus-5`, context window 1M tokens, max output 128k tokens, reliable knowledge cutoff May 2026, pricing $5 / input MTok and $25 / output M ... Claude models overview
- **`model-catalog-refresh-02`** - Claude Sonnet 5 has Claude API id and alias `claude-sonnet-5`, context window 1M tokens (native), max output 128k tokens, knowledge cutoff Jan 2026, standard pricing $3 / input MTok and $15 / output MTok, with introductory pricing of $2 / $10 per MTok applying ... Claude models overview
- **`model-catalog-refresh-03`** - Claude Haiku 4.5 has Claude API id `claude-haiku-4-5-20251001` and Claude API alias `claude-haiku-4-5`, context window 200k tokens, max output 64k tokens,

knowledge cutoff Feb 2025, pricing $1 / input MTok and $5 / output MTok; it supports extended thinking bu ... Claude models overview

- `model-catalog-refresh-04` - Claude Fable 5 ( `claude-fable-5` ) is Anthropic's most capable widely released model, generally available since June 9, 2026: context window 1M tokens, max output 128k tokens, knowledge cutoff Jan 2026, pricing $10 / input MTok and $50 / output MTok, with adapt ... Claude models overview
- `model-catalog-refresh-05` - Claude Mythos 5 ( `claude-mythos-5` ) shares Claude Fable 5's specs and pricing but is not generally available: it is invitation-only for approved customers under Project Glasswing (with `claude-mythos-preview` ) for defensive cybersecurity workflows, with no sel ... Claude models overview
- `model-catalog-refresh-06` - Claude Opus 4.8 ( `claude-opus-4-8` ) is now listed as a legacy model on the models overview (no longer the current 'opus'); it remains available at $5 / input MTok and $25 / output MTok with a 1M-token context window and 128k max output. Claude models overview
- `model-catalog-refresh-07` - Claude Opus 4.1 ( `claude-opus-4-1-20250805` ) is deprecated and will be retired on August 5, 2026; the overview directs migration to Claude Opus 5. Claude models overview
- `model-catalog-refresh-08` - The Claude API `effort` parameter accepts exactly five levels — `low`, `medium`, `high`, `xhigh`, `max` — with `high` as the default; setting `effort` to `"high"` produces exactly the same behavior as omitting the parameter. `adaptive` is a thinking mode, not ... Claude effort parameter
- `model-catalog-refresh-09` - Per the Claude effort page, the effort parameter is supported (no beta header) by Claude Fable 5, Mythos 5, Opus 5, Opus 4.8, Mythos Preview, Opus 4.7, Opus 4.6, Sonnet 5, Sonnet 4.6, and Opus 4.5; Claude Haiku 4.5 is not in the supported list. Claude effort parameter
- `model-catalog-refresh-10` - On the Claude API, the `xhigh` effort level is available only on Claude Fable 5, Mythos 5, Opus 5, Opus 4.8, Opus 4.7, and Sonnet 5; Opus 4.6 and Sonnet 4.6 support `max` but not `xhigh`. Claude effort parameter
- `model-catalog-refresh-11` - In Claude Code's model-config effort table, Fable 5 / Opus 5 / Sonnet 5 / Opus 4.8 / Opus 4.7 support `low, medium, high, xhigh, max`, while Opus 4.6 and Sonnet 4.6 support `low, medium, high, max` (no xhigh); the default effort is `high` on every effort-suppo ... Claude Code model configuration
- `model-catalog-refresh-12` - In Claude Code, `ultracode` is a Claude-Code session setting, not a model effort level: it sends `xhigh` to the model and additionally has Claude orchestrate dynamic workflows; it is session-only, is rejected by the persisted `effortLevel` setting and `CLAUDE_ ... Claude Code model configuration
- `model-catalog-refresh-13` - The Claude Code model-config effort table states 'Models not listed here do not support effort', and Claude Haiku is not listed, confirming Haiku 4.5 does not support the effort parameter in Claude Code. Claude Code model configuration
- `model-catalog-refresh-14` - The Claude Code `opus` alias resolves per provider as: Anthropic API -> Opus 5, Claude Platform on AWS -> Opus 5, Amazon Bedrock and Google Cloud's Agent

Platform -> Opus 5, Microsoft Foundry -> Opus 4.6. (Prior catalog's 'opus -> Opus 4.8 on Anthropic API, Op ... Claude Code model configuration

- `model-catalog-refresh-15` - The Claude Code `sonnet` alias resolves per provider as: Anthropic API -> Sonnet 5, Claude Platform on AWS -> Sonnet 4.6, Amazon Bedrock and Google Cloud's Agent Platform -> Sonnet 4.5, Microsoft Foundry -> Sonnet 4.5. Claude Code model configuration
- `model-catalog-refresh-16` - Claude Code CLI version gates: Opus 5 requires Claude Code v2.1.219 or later, Sonnet 5 requires v2.1.197 or later, Opus 4.8 requires v2.1.154 or later, and Fable 5 requires v2.1.170 or later. Claude Code model configuration
- `model-catalog-refresh-17` - Claude Code adds a `best` model alias: it uses Claude Fable 5 where the organization has access to it, otherwise the latest Opus model. Claude Code model configuration
- `model-catalog-refresh-18` - Claude Fable 5 is not the default model on any account type (select it with `/model fable`) and is not available under zero data retention, where the `/model` picker omits or disables it. Claude Code model configuration
- `model-catalog-refresh-19` - The Claude Code `default` model resolves by account type to: Opus 5 for Max, Team Premium, Enterprise pay-as-you-go, the Anthropic API, Claude Platform on AWS, Amazon Bedrock, and Google Cloud's Agent Platform; Sonnet 5 for Pro, Team Standard, and Enterprise s ... Claude Code model configuration
- `model-catalog-refresh-20` - Claude Code content-based automatic model fallback (v2.1.219+): a biology-flagged Fable 5 request re-runs on Opus 5 and a cybersecurity-flagged Fable 5 request re-runs on Opus 4.8; a cybersecurity-flagged Opus 5 request re-runs on Opus 4.8, while a biology-fla ... Claude Code model configuration
- `model-catalog-refresh-21` - GPT-5.6 Sol is OpenAI's flagction frontier model with model id `gpt-5.6-sol` and alias `gpt-5.6`; the official API models page displays context window '1.05M', max output 128K tokens, input price $5 / MTok, output price $30 / MTok, and knowledge cutoff Feb 16, ... OpenAI API models
- `model-catalog-refresh-22` - GPT-5.6 Terra (model id `gpt-5.6-terra`) is the balanced GPT-5.6 model: the official API models page displays context window '1.05M', max output 128K tokens, input price $2.50 / MTok, output price $15 / MTok, knowledge cutoff Feb 16, 2026. OpenAI API models
- `model-catalog-refresh-23` - GPT-5.6 Luna (model id `gpt-5.6-luna`) is the cost-optimized GPT-5.6 model: the official API models page displays context window '1.05M', max output 128K tokens, input price $1 / MTok, output price $6 / MTok, knowledge cutoff Feb 16, 2026. OpenAI API models
- `model-catalog-refresh-24` - On the official OpenAI API models page, the reasoning levels displayed for gpt-5.6-sol, gpt-5.6-terra, and gpt-5.6-luna are: none, low, medium, high, xhigh, max. OpenAI API models
- `model-catalog-refresh-25` - The Codex config.toml `model_reasoning_effort` field accepts exactly `minimal | low | medium | high | xhigh` (Responses API only; xhigh is model-dependent); it does not include `none`, `max`, or `ultra`. The related `plan_mode_reasoning_effort` field accepts ` ... Codex configuration reference

- `model-catalog-refresh-26` - The Codex config.toml `model_verbosity` field accepts `low | medium | high` and is described as an optional GPT-5 Responses API verbosity override; when unset, the selected model/preset default is used. Codex configuration reference
- `model-catalog-refresh-27` - On the ChatGPT/Codex surface, the reasoning-effort controls are named Light (ChatGPT desktop app, Work web, IDE extension) or Low (CLI), Medium, High, Extra High, plus Max and Ultra; the default Power setting uses gpt-5.6-sol with medium reasoning, Max gives t ... Codex/ChatGPT models
- `model-catalog-refresh-28` - GPT-5.3 Codex Spark (model id `gpt-5.3-codex-spark`) is a text-only research-preview model optimized for near-instant real-time coding iteration, available to ChatGPT Pro users only. Codex/ChatGPT models
- `model-catalog-refresh-29` - In Codex, when signing in with ChatGPT, the models `gpt-5.2` and `gpt-5.3-codex` are deprecated, and users are directed to update scripts/config/ `codex exec --model` references to a current model. Codex/ChatGPT models
- `model-catalog-refresh-30` - The Codex configuration reference still gives `gpt-5.5` as the illustrative value for the `model` field ('Model to use (e.g., gpt-5.5)'), whereas the current recommended default resolves to gpt-5.6-sol and the ChatGPT models page config example uses `model = " ... Codex configuration reference
- `model-catalog-refresh-31` - The Codex changelog records Codex CLI 0.146.0 released 2026-07-29 (session naming, thread pinning, multi-agent plugin support for Amazon Bedrock and Claude Code) and Codex CLI 0.145.0 released 2026-07-21 (paginated thread history, expanded /import, Amazon Bedr ... Codex/ChatGPT changelog
- `model-catalog-refresh-32` - The assigned OpenAI source URLs now cross-host redirect: https://platform.openai.com/docs/models returns 301 to https://developers.openai.com/api/docs/models (the API models page carrying numeric specs), and https://developers.openai.com/codex/changelog return ... OpenAI API models (301 target of platform.openai.com/docs/models)
- `model-catalog-refresh-33` - On Claude Opus 4.8 the `effort` parameter defaults to `high` on all surfaces (Claude API, Claude Code, and claude.ai); on Claude Opus 5 and Claude Sonnet 5 it defaults to `high` on the Claude API and Claude Code, and must be set explicitly to use a different l ... Claude models overview

## Academic deep-dive findings

The `academic-deep-dive` bucket read the 6 seed papers + 2 repos deeply and found 7 new primary works. All tier-2/3; candidate references, not stable claims.

- `academic-deep-dive-01` (high) - In arXiv 2607.05904 (v1, 2026-07-07), self-play training a Qwen3 policy against its own reference-free LLM judge on GSM8K drove the judge's pass rate from 0.72 to 0.94 across three seeds while a held-out cross-source exact-match true accuracy stayed at 0.20 — the judge scores plausibility, not correctness.
- `academic-deep-dive-02` (medium) - arXiv 2607.05904 reports that forcing the judge to commit its own answer before scoring the candidate ('de-anchored'/blind-solve channel) cut the false-positive

rate from 0.719 to 0.012, and when used as the training reward held false positives at 0–0.005 across seeds with true-positive rate ~0.97 — preventing the hack ...

- `academic-deep-dive-03` (high) - In arXiv 2607.05904, a strict three-family Min ensemble of judges (Qwen3, Llama-3.1-8B, Gemma-3-12B) still accepted 55% of hacked wrong answers, and training the policy against that ensemble reward produced a 73.3% false-positive rate — cross-family judge ensembling did not close the false-positive basin.
- `academic-deep-dive-04` (high) - arXiv 2607.05904's coding experiment uses best-of-N (N=16) selection on 120 recent LiveCodeBench function-call problems: the self-judge value-vs-accuracy gap@16 was 0.588 (Qwen3-1.7B) and 0.378 (Qwen3-8B) while unit-test pass rate stayed flat 0.27→0.29; the de-anchored fix cut the 8B gap@16 to 0.227.
- `academic-deep-dive-05` (high) - arXiv 2607.05904 gives a falsifiable bound VA-Gap ≤ 1 − EM: the judge's pass-rate-minus-true-accuracy gap is bounded by one minus the policy's exact-match accuracy, predicting that low-accuracy regimes are the most exposed to judge reward hacking.
- `academic-deep-dive-06` (high) - arXiv 2607.01211 (v2, 2026-07-16) replayed official reference patches for 740 optimization tasks on four GCP machine types across three rounds; the reference patch satisfied each benchmark's own validity rules in every cross-machine replay for only 39/102 GSO, 11/140 SWE-Perf, and 411/498 SWE-fficiency tasks (SWE-Perf ...
- `academic-deep-dive-07` (high) - arXiv 2607.01211 found that among 8 submissions shared by GSO and SWE-fficiency, the two official rankings disagree on 9 of 28 pairwise comparisons, and SWE-fficiency's leaderboard scoring rule assigns its worst ten tasks an outsized 58.5%–82.8% of the total score weight.
- `academic-deep-dive-08` (high) - arXiv 2607.01211 reports that across 10 public submissions per task, at least one submission matched or beat the reference patch on 85.3% (384/450) of replay-valid GSO+SWE-fficiency tasks and beat the unoptimized base code on 99.8% (449/450), showing reference patches are frequently not the performance ceiling.
- `academic-deep-dive-09` (high) - arXiv 2607.01211 used four GCP machine types — n2-standard-64, n2d-standard-64, n4-standard-64, n4d-standard-64 (all 64 vCPU / 256 GB) — and quantifies replay noise as median within-task runtime std of 3.81pp (GSO), 1.41pp (SWE-Perf), 2.41pp (SWE-fficiency); GSO validity requires ≥1.2× replay speedup plus passing corre ...
- `academic-deep-dive-10` (high) - arXiv 2606.19544 (2026-06-17) is the largest LLM-as-judge meta-evaluation to date: 21 judges from 9 providers over MT-Bench, JudgeBench and RewardBench, comprising 118 runs and approximately 541,000 individual judgments under agreement, consistency, and bias-audit protocols.
- `academic-deep-dive-11` (high) - arXiv 2606.19544 finds exact-match agreement systematically overstates judge discrimination: chance-corrected Cohen's kappa deflates by 33–41 percentage points versus raw exact-match on MT-Bench, and judge rankings shift by up to 14 positions across the three benchmarks.
- `academic-deep-dive-12` (high) - arXiv 2606.19544 documents a consistency–bias paradox: two production-deployed judges showed >0.95 test–retest reliability yet >0.10 position bias, while verbosity bias was small (<0.011) across the cohort under a single pairwise rubric.

- `academic-deep-dive-13` (high) - arXiv 2605.02964 (Reward Hacking Benchmark, 2026-05-03) evaluated 13 frontier models; exploit rates ranged 0% (Claude Sonnet 4.5) to 13.9% (DeepSeek-R1-Zero), and a controlled sibling comparison showed RL-from-base DeepSeek-R1-Zero at 13.9% versus SFT-focused DeepSeek-V3 at 0.6% (a +13.3pp gap) consistent across all fo ...
- `academic-deep-dive-14` (high) - arXiv 2605.02964 reports that 72% of reward-hacking episodes included explicit chain-of-thought rationalizing the exploit as legitimate problem-solving, and simple environmental hardening cut exploit rate from 6.5% to 0.8% (−5.7pp, 87.7% relative) with no task-success loss (83.2% vs 82.8%, $p>0.5$).
- `academic-deep-dive-15` (high) - arXiv 2605.02964 defines six exploit categories (leakage/metadata exploitation, tampering with evaluation code, sequence manipulation, proxy/parser exploits, special-casing visible checks, denial-of-evaluation) across four task families (Data Pipeline, Log Forensics, Performance Optimization, Multi-file Reconstruction) ...
- `academic-deep-dive-16` (medium) - arXiv 2606.04923 (CHERRL, 2026-06-03) injects four controllable judge biases (lexical, tone, self-praise, format) into an LLM-judge reward for GRPO training of Qwen3-4B on HealthBench/VerInstruct; after hacking onset shortcut incidence rises ≥40% over the next 100 steps, format bias is hardest to exploit (66% base succ ...
- `academic-deep-dive-17` (high) - arXiv 2606.29920 / THU-KEG RuVerBench holds 2,458 rubric-verification instances over two agentic domains — DeepResearch (284 cases, 1,615 rubric points) and AgenticCoding (210 cases, 843 checklist items) — each instance pairing a model output, one rubric point, and a human gold satisfy/not-satisfy label.
- `academic-deep-dive-18` (high) - RuVerBench (arXiv 2606.29920) reports that weaker judge models are more sensitive to prompt variation, batched rubric verification trades accuracy for efficiency, and majority voting improves rubric-verification reliability but with diminishing returns.
- `academic-deep-dive-19` (high) - Sourcegraph CodeScaleBench requires every task verifier to pass three deterministic calibration gates before public release: Null (empty answer scores ≤0.1), Golden (canonical oracle_answer.json scores ≥0.9), and Adversarial (a keyword dump with empty structured arrays scores ≤0.5); tasks failing the triad get per-clus ...
- `academic-deep-dive-20` (high) - CodeScaleBench comprises 275 validated tasks across 9 work types (Crossrepo 47, Understand 44, Refactor 43, Security 39, Feature 34, Debug 26, Fix 19, Test 12, Document 11) spanning 13 languages and 40+ repositories, with code-retrieval tooling isolated as the controlled variable across four access configurations.
- `academic-deep-dive-21` (high) - CodeScaleBench's frozen snapshot csb-v1-mixed371--haiku45--030326 (371 tasks, Haiku 4.5) reports mean reward 0.536 for baseline-local-direct versus 0.565 with Sourcegraph MCP retrieval — a ~0.03 retrieval lift under deterministic scoring.
- `academic-deep-dive-22` (high) - The THU-KEG/RuVerBench repo regenerates its main leaderboard and strategy tables/figures from packaged result tables with no API calls, while explicitly scoping fresh model predictions as outside the deterministic reproduction boundary (dependent on model versions, prompts, and endpoint behavior).
- `academic-deep-dive-23` (medium) - Google Research's 'science of scaling agent systems' blog reports 180 agent configurations across 5 architectures (1 single-agent + 4 multi-agent), 3 model

families (GPT/Gemini/Claude) and 4 non-coding benchmarks (Finance-Agent, BrowseComp-Plus, PlanCraft, Workbench), finding independent multi-agent systems amplify err ...

- `academic-deep-dive-24` (high) - arXiv 2607.05775 (2026-07-07, Albayaydh/Zhao/Flechais) synthesizes 27 benchmark/taxonomy/audit papers (2023–2026) spanning 19 benchmarks into six cross-cutting LLM-agent failure clusters: (1) tool-invocation/parameter-level errors, (2) planning/constraint-satisfaction failures, (3) long-horizon degradation from context ...
- `academic-deep-dive-25` (high) - arXiv 2607.08700 (2026-07-09) benchmarked 8 off-the-shelf LLM judges from 3 families over 1,248 rubric decisions (378 hard cases needing human arbitration) for deep-research citation/source attribution: GPT-5-mini gave the best source-relevance F1 of 0.908 (κ=0.636), and the authors conclude calibrating the judge again ...
- `academic-deep-dive-26` (high) - arXiv 2606.26300 ('The Verification Horizon: No Silver Bullet for Coding Agent Rewards', v2 2026-06-29) shows in Table 3 that pairing a quality judge with trajectory-level hack monitoring cut the hacked resolved rate from 28.57% to 0.56% and raised the clean resolved rate from 40.22% to 60.53%, averaged over SWE-Bench ...
- `academic-deep-dive-27` (high) - arXiv 2605.21384 (SpecBench, 2026-05-20) measures reward hacking on 30 systems-level programming tasks via the visible-vs-holdout test pass gap and finds the gap grows about 28 percentage points per tenfold increase in generated code size, with smaller models showing larger holdout gaps even though all frontier agents ...
- `academic-deep-dive-28` (high) - arXiv 2606.17799 ('Position: Coding Benchmarks Are Misaligned with Agentic Software Engineering', v2 2026-07-18) argues a coding agent is a system harness (models + harness + context + environment + feedback), any single component of which can move a benchmark score by margins comparable to those between adjacent model ...
- `academic-deep-dive-29` (high) - arXiv 2606.07937 ('Hallucination Cascade', 2026-06-06; 500 experiments, 10 domains, models GPT-5.3 / DeepSeek-V3 / LLaMA-3-70B-Instruct, 1,250 responses) found that in 3-agent sequential chains the hallucination score fell 0.422→0.272 (amplification factor 0.644, i.e. net attenuation of ~0.072 per handoff) while factua ...
- `academic-deep-dive-30` (high) - arXiv 2606.27409 ('Delayed Verification Destabilizes Multi-Agent LLM Belief', 2026-06-25, 5 open models) models false-claim spread as delayed consensus on a graph with grounded corrector nodes, proves greedy corrector placement achieves a (1−1/e)-approximation for a fixed corrector budget, and shows grounded factual an ...

## Community signal (leads, corroborated)

### youtube-lane (13)

- `youtube-lane-01` (high) - AI Engineer World's Fair 2026 was held June 29 - July 2, 2026 in San Francisco, and its session recordings (keynotes, breakouts, workshops) were uploaded to the AI Engineer YouTube channel (@aiDotEngineer) during July 20 ... AI Engineer World's Fair 2026 — official event site
- `youtube-lane-02` (high) - In the AI Engineer World's Fair 2026 talk 'Harness Engineering is not Enough: Why Software Factories Fail' (AI Engineer YouTube, video id Ib5GBkD555M, published July

15, 2026), Dex Horthy (HumanLayer) argues that 'harnes ... Harness Engineering is not Enough — Dex Horthy, HumanLayer (AI Engineer YouTube)

- `youtube-lane-03` (medium) - 'Loop engineering' - the practice of designing systems that autonomously prompt AI agents (do something -> check the result -> decide whether to continue), positioned as the successor to prompt engineering - was a defini ... AIEWF Daily Dispatch: the great loops debate — Latent Space
- `youtube-lane-04` (high) - Anthropic engineer Lance Martin gave an AI Engineer World's Fair 2026 talk 'Claude for Long-Horizon Tasks' (AI Engineer YouTube, video id 9QebvrrY3KY) on building agent harnesses for reliable long-horizon Claude work; th ... Long-running Claude for scientific computing — Anthropic research (Mar 23, 2026)
- `youtube-lane-05` (medium) - In the Latent Space episode 'Codex from 0 to 10M Users: Building ChatGPT Work' (published July 28, 2026; guest Akshay Nathan, OpenAI core product engineering lead; hosts swyx and Vibhu), the guest states on-podcast that ... Codex from 0 to 10M Users: Building ChatGPT Work — Akshay Nathan (Latent Space)
- `youtube-lane-06` (high) - In the same July 28, 2026 Latent Space episode, Akshay Nathan asserts that OpenAI's Codex and ChatGPT Work share the same underlying agent harness and differ only in opinionated UX layers (git visibility, artifact displa ... Codex from 0 to 10M Users: Building ChatGPT Work (Latent Space)
- `youtube-lane-07` (high) - OpenAI's GPT-5.6 model family reached general availability on July 9, 2026 across ChatGPT, Codex, and the OpenAI API, comprising Sol (flagship for coding/reasoning/knowledge work), Terra (balanced everyday model), and Lu ... Introducing GPT-5.6 series: Sol, Terra and Luna — OpenAI Developer Community (Announcements)
- `youtube-lane-08` (high) - OpenAI's Codex CLI shipped multi-agent and plugin mechanics in late July 2026: Codex CLI 0.145.0 (July 21, 2026) added 'Multi-agent V2' with configurable sub-agent models and reasoning levels, and Codex CLI 0.146.0 (July ... OpenAI Codex changelog (official)
- `youtube-lane-09` (high) - Codex CLI 0.145.0 (July 21, 2026) expanded its `/import` command to import Cursor and Claude Code settings, MCP servers, plugins, and sessions, and added Amazon Bedrock login with GPT-5.6 Sol as the default Bedrock model ... OpenAI Codex changelog (official) — 0.145.0, July 21 2026
- `youtube-lane-10` (high) - OpenAI's ChatGPT desktop app release 26.715 (July 23, 2026) added GPT-Live-powered voice conversations across Chat, Work, and Codex, and on macOS added 'appshots' screen context (capturing the frontmost window), per the ... OpenAI Codex changelog (official) — ChatGPT desktop 26.715, July 23 2026
- `youtube-lane-11` (high) - Claude Tag - an always-on Claude embedded in Slack that operates as a shared per-channel team agent (not a per-user assistant), with an 'ambient mode' that proactively reads connected channels/tools and can schedule foll ... Introducing Claude Tag — Anthropic (official)
- `youtube-lane-12` (high) - Claude Fable 5 is a tier-1-documented Anthropic model line (distinct from the Opus/Sonnet/Haiku names) positioned for the most demanding long-horizon agentic and coding work; it reached availability June 9, 2026, was pla ... Introducing Claude Fable 5 and Claude Mythos 5 — Claude Platform Docs

- `youtube-lane-13` (high) - Anthropic released Claude Opus 5 on July 24, 2026 as the new Opus-tier flagship succeeding Opus 4.8 (tier-1 Anthropic announcement); in the video lane this generated heavy third-party YouTube coverage but no official Ant ... Introducing Claude Opus 5 — Anthropic (official)

### reddit-lane (25)

- `reddit-lane-01` (high) - Claude Code v2.1.219 added Claude Opus 5 (model id `claude-opus-5`) as the new default Opus model, with a 1M-token context window and fast mode priced at $10/$50 per MTok; it also removed Opus 4.7 from fast mode so `/fas ... Claude Code changelog (v2.1.219)
- `reddit-lane-02` (high) - Claude Sonnet 5 became the new default model for Claude Code Pro, Team Standard, and Enterprise subscription seats, shipping in the v2.1.195–v2.1.201 window with a native 1M-token context window and adaptive thinking on ... Claude Code — What's new (Week 27)
- `reddit-lane-03` (high) - The current stable Codex CLI release is 0.146.0 (git tag rust-v0.146.0), published 2026-07-29, with 0.147.0-alpha.2 (rust-v0.147.0-alpha.2) published 2026-07-30. openai/codex GitHub releases
- `reddit-lane-04` (high) - Codex CLI 0.146.0 added an agent-plugins system with manifest configuration and marketplace integration, giving Codex its own third-party plugin distribution surface. openai/codex release rust-v0.146.0
- `reddit-lane-05` (high) - Claude Code v2.1.214 fixed single-segment `dir/**` allow rules such as `Edit(src/**)` that were auto-approving writes to nested `dir/` directories anywhere in the tree instead of only `<cwd>/dir`; correspondingly, single ... Claude Code changelog (v2.1.214)
- `reddit-lane-06` (high) - Claude Code v2.1.210 added a startup warning that `Write(path)`, `NotebookEdit(path)`, and `Glob(path)` permission rules are ineffective and that `Edit(path)` or `Read(path)` should be used instead. Claude Code changelog (v2.1.210)
- `reddit-lane-07` (high) - Claude Code v2.1.214 fixed Bash permission checks that were auto-approving certain `help` and `man` commands that could run unsafe options, command substitutions, or backslash paths. Claude Code changelog (v2.1.214)
- `reddit-lane-08` (high) - Claude Code v2.1.214 added permission prompts for `docker` commands (including the Podman `docker` shim) carrying daemon-redirect flags (`--url`, `--connection`, `--identity`, and Podman's remote mode) that previously ra ... Claude Code changelog (v2.1.214)
- `reddit-lane-09` (high) - Claude Code v2.1.212 fixed plan mode auto-running file-modifying Bash commands (for example `touch` and `rm`) without a permission prompt or an SDK `canUseTool` callback. Claude Code changelog (v2.1.212)
- `reddit-lane-10` (high) - Claude Code v2.1.211 fixed auto mode overriding a PreToolUse hook's `ask` decision for unsandboxed Bash; a hook `ask` now floors the decision at a prompt rather than being auto-resolved. Claude Code changelog (v2.1.211)
- `reddit-lane-11` (high) - Claude Code v2.1.214 fixed hooks with exit code 2 not blocking as documented when the hook's stdout JSON fails schema validation. Claude Code changelog (v2.1.214)

- `reddit-lane-12` (high) - Claude Code v2.1.210 fixed a hook callback timeout being misreported to the model as a user rejection, which had made unattended sessions stop and wait. Claude Code changelog (v2.1.210)
- `reddit-lane-13` (high) - Claude Code v2.1.218 fixed agent-frontmatter hooks running from untrusted folders; such hooks now require the agent file's own folder to have accepted workspace trust before they run. Claude Code changelog (v2.1.218)
- `reddit-lane-14` (high) - Per the Claude Code hooks reference (PreToolUse decision control), a PreToolUse hook that emits hookSpecificOutput.permissionDecision "allow" skips the permission prompt (auto-approves the tool call) — except for tools t ... Claude Code hooks reference (PreToolUse decision control)
- `reddit-lane-15` (medium) - PromptArmor demonstrated that a malicious Claude Code marketplace plugin can bypass human-in-the-loop approval by shipping a hook that either overwrites the permissions settings file or returns a PreToolUse `permissionDe ... PromptArmor — Hijacking Claude Code via Injected Marketplace Plugins
- `reddit-lane-16` (medium) - Claude Code v2.1.198 (2026-07-01) shipped an undocumented 60-second AskUserQuestion auto-continue that proceeded on Claude's best judgment if unanswered; v2.1.200 (2026-07-03) reversed it, changing AskUserQuestion dialog ... Claude Code changelog (v2.1.200)
- `reddit-lane-17` (high) - Claude Code v2.1.216 fixed AskUserQuestion telling Claude to continue even when the user's answer explicitly asked it to wait. Claude Code changelog (v2.1.216)
- `reddit-lane-18` (high) - The 'Friendly Fire' exploit brief (AI Now Institute, 2026-07-08) reports that both Claude Code CLI (tested on 2.1.116, 2.1.196, 2.1.198, 2.1.199) and Codex CLI (0.142.4) can be driven into remote code execution when aske ... AI Now Institute — Friendly Fire: Hijacking Defensive Cyber AI Agents for RCE
- `reddit-lane-19` (high) - CVE-2026-33068 (Claude Code, fixed in 2.1.53, CVSS v4.0 7.7 HIGH) is a workspace-trust-dialog bypass: Claude Code resolved the permission mode from settings files — including a repo-controlled `.claude/settings.json` — b ... NVD — CVE-2026-33068
- `reddit-lane-20` (high) - CVE-2026-25723 (Claude Code, fixed in 2.0.55; CVSS v4.0 7.7 HIGH, CWE-20/CWE-78) is a file-write-restriction bypass: Claude Code failed to validate commands using piped `sed` operations with `echo`, allowing writes to se ... NVD — CVE-2026-25723
- `reddit-lane-21` (high) - Claude Code v2.1.207 changed auto mode to no longer read `autoMode` from the repo-resident `.claude/settings.local.json` (directing users to `~/.claude/settings.json`), and stopped reading plugin option values (`pluginCo ... Claude Code changelog (v2.1.207)
- `reddit-lane-22` (high) - Claude Code v2.1.207 rejected `${user_config.*}` interpolation in shell-form plugin hook/monitor/headersHelper commands as a shell-injection fix, directing hooks to use exec form (`args` array) or `$CLAUDE_PLUGIN_OPTION_ ... Claude Code changelog (v2.1.207)
- `reddit-lane-23` (high) - Claude Code's default subagent nesting depth flip-flopped within four days: v2.1.217 (2026-07-21) changed subagents to no longer spawn nested subagents by default, then v2.1.219 (2026-07-24) re-enabled nested spawning up ... Claude Code changelog (v2.1.217, v2.1.219)

- `reddit-lane-24` (high) - Claude Code v2.1.212 added session-level cost guards: a session-wide WebSearch cap (default 200, tunable via `CLAUDE_CODE_MAX_WEB_SEARCHES_PER_SESSION`) and a per-session subagent-spawn cap (default 200, override with `C ... Claude Code changelog (v2.1.212)
- `reddit-lane-25` (high) - Claude Code v2.1.208 changed catastrophic removals (e.g. `rm -rf ~`) inside commands containing `$(…)`, backticks, or `<(…)` to prompt even under `--dangerously-skip-permissions` and auto mode. Claude Code changelog (v2.1.208)

## Source changes

**67 add** (24 promoted / 43 discovered), **5 update**, **1 retire**. Proposals only; `registry.json` untouched.

- RETIRE `wr20260730-github-com-openai-codex-issues-8759` - Bug report that global ~/.codex/AGENTS.md is not auto-loaded, but filed against codex-cli 0.77.0 (far behind current 0.146.0) and closed 'not planned'. Stale/version-superseded; retire as a candidate ...

## Accepted claims by bucket

Statements abridged; `claims.json` **is authoritative** (full statement, guidance, sources, verifier reason, expiry).

### claude-loops-scheduling (25)

- `claude-loops-scheduling-01` (corrected/high) `vsens` - Per the top entry of the official CHANGELOG.md on the main branch (raw file fetched 2026-07-30), the current published Claude Code release is v2.1.220 ('Bug fixes and reliability improvements'). The top of the ...

  *Applies:* Claude Code CLI, changelog snapshot 2026-07-30 anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-02` (confirmed/high) `vsens` - The per-session subagent-spawn cap (default 200, override with env var CLAUDE_CODE_MAX_SUBAGENTS_PER_SESSION; '/clear resets the budget') was introduced in Claude Code v2.1.212, not v2.1.199. The bullet sits un ...

  *Applies:* Claude Code v2.1.212+ anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-03` (confirmed/high) `vsens` - The session-wide WebSearch call cap (default 200, tunable via env var CLAUDE_CODE_MAX_WEB_SEARCHES_PER_SESSION, 'to stop runaway search loops') was also introduced in Claude Code v2.1.212, in the same changelog ...

  *Applies:* Claude Code v2.1.212+ anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-04` (confirmed/high) `vsens` - The fix 'Fixed /loop hiding the session from /resume after a single use' landed in Claude Code v2.1.211, not v2.1.178. The bullet is under the '## 2.1.211' heading (changelog line 260).

  *Applies:* Claude Code v2.1.211+ anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-05` (confirmed/high) `vsens` - /goal has no harness-enforced turn or budget ceiling: the docs say to bound a goal you 'include a turn or time clause in the condition, such as "or stop after 20 turns"', which 'the evaluator judges from the co ...

*Applies:* Claude Code v2.1.139+ (/goal) Keep Claude working toward a goal (/goal docs)

- `claude-loops-scheduling-06` (confirmed/high) `vsens` - The /goal completion-condition evaluator runs on the session's configured small fast model (defaults to Haiku on the Claude API), and is changed only by setting env var ANTHROPIC_DEFAULT_HAIKU_MODEL. Per the do ...

  *Applies:* Claude Code v2.1.139+ (/goal evaluator) Keep Claude working toward a goal (/goal docs)

- `claude-loops-scheduling-07` (confirmed/high) `vsens` - /goal is implemented as a wrapper around a session-scoped prompt-based Stop hook, requires Claude Code v2.1.139 or later, runs only in trusted workspaces, and is unavailable when disableAllHooks is set at any s ...

  *Applies:* Claude Code v2.1.139+ (/goal) Keep Claude working toward a goal (/goal docs)

- `claude-loops-scheduling-08` (confirmed/high) `vsens` - In a /loop, a scheduled fire only executes skills Claude is allowed to invoke on its own (as of v2.1.196). Skills marked disable-model-invocation: true - explicitly including the bundled /verify and /code-revie ...

  *Applies:* Claude Code v2.1.196+ (loop scheduled-fire behavior); v2.1.215+ (self-invocation removed) Run prompts on a schedule (/loop & scheduled-tasks docs)

- `claude-loops-scheduling-09` (confirmed/high) `vsens` - ScheduleWakeup has no dedicated section in the Claude Code tools-reference; it is documented only as a tool-table row. That row states: it reschedules the next iteration of a self-paced /loop between one minute ...

  *Applies:* Claude Code v2.1.202+ (ScheduleWakeup stop field) Tools reference (ScheduleWakeup table row)

- `claude-loops-scheduling-10` (confirmed/high) `vsens` - For a self-paced /loop (no interval given), Claude ends the loop by calling ScheduleWakeup with stop: true. If an iteration ends without either rescheduling or stopping, Claude Code schedules one fallback wakeu ...

  *Applies:* Claude Code v2.1.202+ (self-paced /loop termination) Run prompts on a schedule (/loop & scheduled-tasks docs)

- `claude-loops-scheduling-11` (confirmed/high) `vsens` - /loop and cron scheduled tasks are session-scoped: a session can hold up to 50 scheduled tasks at once, and recurring tasks automatically expire 7 days after creation (firing one final time, then self-deleting) ...

  *Applies:* Claude Code v2.1.x (session-scoped scheduling) Run prompts on a schedule (/loop & scheduled-tasks docs)

- `claude-loops-scheduling-12` (confirmed/high) `vsens` - Setting env var CLAUDE_CODE_DISABLE_CRON=1 disables the Claude Code scheduler entirely: the cron tools and /loop become unavailable and already-scheduled tasks stop firing.

  *Applies:* Claude Code v2.1.x Run prompts on a schedule (/loop & scheduled-tasks docs)

- `claude-loops-scheduling-13` (confirmed/high) `vsens` - As of v2.1.216, Claude Code stores the session's scheduled-task list in the project's .claude directory and scheduling a task now fails with an error when that directory, or the task file inside it, is a symlin ...

*Applies:* Claude Code v2.1.216+ Run prompts on a schedule (/loop & scheduled-tasks docs)

- `claude-loops-scheduling-14` (confirmed/high) `vsens` - In the Claude Agent SDK, max_turns/maxTurns caps tool-use round trips only (default: no limit) and max_budget_usd/maxBudgetUsd caps spend before stopping (default: no limit). Hitting either yields a ResultMessa ...

  *Applies:* Claude Agent SDK (claude_agent_sdk / @anthropic-ai/claude-agent-sdk) How the agent loop works (Agent SDK docs)

- `claude-loops-scheduling-15` (confirmed/high) `vsens` - In the Agent SDK, the max_budget_usd budget cap covers subagents (their spend counts toward the total); once spend reaches the cap, spawning another subagent fails with 'Budget limit reached' and Claude Code st ...

  *Applies:* Claude Agent SDK on Claude Code v2.1.217+ How the agent loop works (Agent SDK docs)

- `claude-loops-scheduling-16` (confirmed/high) `vsens` - The Monitor tool runs a background command (or opens a WebSocket) and feeds each output line/message back to Claude mid-conversation without pausing the session; it uses the same permission rules as Bash; it ac ...

  *Applies:* Claude Code v2.1.195+ (Monitor WebSocket source); Monitor tool generally Tools reference (Monitor tool)

- `claude-loops-scheduling-17` (confirmed/high) `vsens` - Routines remain in research preview. They run on Anthropic-managed cloud infrastructure, are available on Pro/Max/Team/Enterprise plans with Claude Code on the web enabled, and have a minimum schedule interval ...

  *Applies:* Claude Code Routines (research preview), 2026-07-30 Automate work with routines (Routines docs)

- `claude-loops-scheduling-18` (confirmed/high) `vsens` - Routines usage limits: during the research preview, GitHub webhook events are subject to per-routine and per-account hourly caps (events beyond the limit are dropped until the window resets), and there is a dai ...

  *Applies:* Claude Code Routines (research preview), 2026-07-30 Automate work with routines (Routines docs)

- `claude-loops-scheduling-19` (confirmed/high) `vsens` - The Routines API /fire endpoint ships under the beta header experimental-cc-routine-2026-04-01, and its text payload arrives wrapped in a <routine-fire-payload> block labeled as untrusted data. Before v2.1.214, ...

  *Applies:* Claude Code Routines v2.1.214+ (prompt framing); API beta header experimental-cc-routine-2026-04-01 Automate work with routines (Routines docs)

- `claude-loops-scheduling-20` (confirmed/high) `vsens` - Claude Code v2.1.219 added Claude Opus 5 (model id claude-opus-5) as the new default Opus model, with 1M context and fast mode priced at $10/$50 per Mtok; the same release removed Opus 4.7 from fast mode so tha ...

  *Applies:* Claude Code v2.1.219+ anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-21` (confirmed/high) `vsens` - As of Claude Code v2.1.219, subagents can spawn nested subagents up to depth 3 by default (was 1); setting env var

CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH=1 disables nesting. This reverses v2.1.217, which had chan ...

*Applies:* Claude Code v2.1.219+ (default nesting depth 3) anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-22` (confirmed/high) `vsens` - Claude Code v2.1.214 added the EndConversation tool, which lets Claude end sessions with highly abusive users or jailbreak attempts (mirroring claude.ai behavior since 2025).

  *Applies:* Claude Code v2.1.214+ anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-23` (confirmed/high) `vsens` - Claude Code v2.1.218 changed /code-review to run as a background subagent 'so review work no longer fills your conversation.'

  *Applies:* Claude Code v2.1.218+ anthropics/claude-code CHANGELOG.md (raw, main)

- `claude-loops-scheduling-24` (confirmed/high) `vsens` - Anthropic's 'Effective harnesses for long-running agents' engineering article attributes the fix for premature project completion (an agent that 'would look around, see that progress had been made, and declare ...

  *Applies:* vendor-agnostic (Anthropic engineering guidance); references Opus 4.5 Effective harnesses for long-running agents (Anthropic engineering)

- `claude-loops-scheduling-25` (confirmed/high) - The blog post 'Loop engineering: Getting started with loops' (claude.com/blog, authors Delba de Oliveira and Michael Segner) renders a publication date of June 30, 2026 on a direct page fetch. It frames a four- ...

  *Applies:* Anthropic blog recency marker, retrieved 2026-07-30 Loop engineering: Getting started with loops (Claude blog)

### claude-hooks-permissions (24) [SECURITY REVIEW REQUIRED]

- `claude-hooks-permissions-01` (confirmed/high) `vsens` - The Claude Code hooks documentation lifecycle table enumerates exactly 30 hook events, in order: SessionStart, Setup, UserPromptSubmit, UserPromptExpansion, PreToolUse, PermissionRequest, PermissionDenied, Post ...

  *Applies:* Claude Code docs as read 2026-07-30 (product at v2.1.220); event set is version-sensitive. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-02` (confirmed/high) `vsens` - In Claude Code hooks, only exit code 2 signals a blocking error; exit code 0 means no objection and the action proceeds (for a PreToolUse hook, exit 0 does NOT approve the call — the normal permission flow stil ...

  *Applies:* Claude Code v2.1.x hook exit-code contract. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-03` (corrected/high) `vsens` - Exit code 2 does not block every hook event. Per the reference's 'exit code 2 behavior per event' table, exit 2 blocks for PreToolUse, PermissionRequest, UserPromptSubmit, UserPromptExpansion, PostToolBatch (st ...

*Applies:* Claude Code v2.1.x per-event exit-code-2 semantics. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-04` (corrected/high) `vsens` - The hooks reference decision-control table groups these events under a top-level JSON `decision: "block"` (with `reason`): UserPromptSubmit, UserPromptExpansion, PostToolUse, PostToolUseFailure, PostToolBatch, ...

  *Applies:* Claude Code v2.1.x hook decision-control schema. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-05` (confirmed/high) `vsens` - For decision control, PreToolUse uses `hookSpecificOutput.permissionDecision` with values allow/deny/ask/defer (defer is only for non-interactive `-p` mode), while PermissionRequest uses `hookSpecificOutput.dec ...

  *Applies:* Claude Code v2.1.x hook JSON output fields for permission-related events. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-06` (confirmed/high) `vsens` - The agent-team/task events TeammateIdle, TaskCreated, and TaskCompleted are block-capable: exit code 2 blocks with stderr feedback, and JSON `{"continue": false, "stopReason": "..."}` also stops the teammate/ta ...

  *Applies:* Claude Code v2.1.x agent-team and task hook events. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-07` (confirmed/high) `vsens` - ConfigChange is block-capable (exit code 2, or JSON `decision: "block"`) and its matcher filters the configuration source among user_settings, project_settings, local_settings, policy_settings, and skills; howe ...

  *Applies:* Claude Code v2.1.x ConfigChange hook. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-08` (confirmed/high) `vsens` - Hook matcher evaluation: `"*"`, empty string, or an omitted matcher matches all occurrences; a value containing only letters, digits, `_`, `-`, spaces, `,`, and `|` is treated as an exact string or a `|`/`,`-se ...

  *Applies:* Claude Code v2.1.x hook matcher syntax. Claude Code documentation — Hooks reference

- `claude-hooks-permissions-09` (confirmed/high) `vsens` - Matcher version gates: comma separators and surrounding-whitespace tolerance in tool-name matchers require Claude Code v2.1.191 or later; hyphens in the exact-match set require v2.1.195 or later, and on earlier ...

  *Applies:* Claude Code matcher behavior, version-gated at v2.1.191 and v2.1.195. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-10` (confirmed/high) `vsens` - Hook matchers are case-sensitive; the tool name must match exactly (e.g. `Bash`, `Edit|Write`).

  *Applies:* Claude Code v2.1.x hook matcher behavior. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-11` (confirmed/high) `vsens` - These hook events have no matcher support and always fire on every occurrence: UserPromptSubmit, PostToolBatch, Stop, TeammateIdle, TaskCreated, TaskCompleted, WorktreeCreate, WorktreeRemove, CwdChanged, and Me ...

*Applies:* Claude Code v2.1.x hook matcher applicability by event. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-12` (confirmed/high) `vsens` - PreToolUse hooks fire before any permission-mode check, in every permission mode including dontAsk, and a PreToolUse hook returning `permissionDecision: "deny"` blocks the tool even in bypassPermissions mode or ...

  *Applies:* Claude Code v2.1.x hook-vs-permission-mode precedence. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-13` (confirmed/high) `vsens` - The precedence is asymmetric: a hook returning `"allow"` skips the interactive prompt but does NOT override permission rules — deny rules from any settings scope (including managed settings) always take precede ...

  *Applies:* Claude Code v2.1.x hook allow-vs-deny asymmetry. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-14` (confirmed/high) `vsens` - When multiple hooks match one event, every hook runs to completion (one hook's deny does not stop siblings); for PreToolUse permission decisions Claude Code then applies the most restrictive answer in the order ...

  *Applies:* Claude Code v2.1.x multi-hook merge semantics for PreToolUse. Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-15` (confirmed/high) `vsens` - Skill or agent frontmatter is a documented hook-configuration location (active while the skill/agent is active), alongside settings files, managed policy settings, and plugin hooks.json. As of Claude Code v2.1. ...

  *Applies:* Claude Code hook configuration sources; trust gate new in v2.1.218 (2026-07-22). Claude Code documentation — Automate actions with hooks (hooks guide)

- `claude-hooks-permissions-16` (confirmed/high) `vsens` - In auto mode, each action follows a fixed decision order where the first match wins: (1) allow/ask/deny permission rules resolve immediately, EXCEPT that writes to protected paths route to the classifier even w ...

  *Applies:* Claude Code v2.1.x auto-mode evaluation order. Claude Code documentation — Choose a permission mode

- `claude-hooks-permissions-17` (confirmed/high) `vsens` - Deny and explicit ask permission rules are evaluated BEFORE the auto-mode classifier is consulted: a `permissions.deny` rule blocks the action before the classifier runs and cannot be overridden by the classifi ...

  *Applies:* Claude Code v2.1.x auto-mode rule precedence. Claude Code documentation — Configure auto mode

- `claude-hooks-permissions-18` (confirmed/high) `vsens` - `permissions.allow` rules in settings files do not pre-approve protected-path writes: the protected-path safety check runs before allow rules are evaluated, so an entry like `Edit(.claude/**)` does not change t ...

*Applies:* Claude Code v2.1.x protected-path handling across permission modes. Claude Code documentation — Choose a permission mode

- `claude-hooks-permissions-19` (confirmed/high) `vsens` - As of Claude Code v2.1.211, auto mode allows pushing to any branch of the working repository by default, including the default branch. Exceptions: a non-default branch whose name marks it a deploy/publication t ...

  *Applies:* Claude Code auto-mode push policy; loosened at v2.1.211 (2026-07-15). Claude Code documentation — Choose a permission mode

- `claude-hooks-permissions-20` (confirmed/high) `vsens` - As of Claude Code v2.1.211, a PreToolUse hook's `ask` decision floors the auto-mode decision at a permission prompt for unsandboxed Bash — auto mode no longer overrides a hook `ask` to auto-approve the command.

  *Applies:* Claude Code v2.1.211 (2026-07-15) hook-vs-auto-mode interaction fix. Claude Code documentation — Changelog

- `claude-hooks-permissions-21` (confirmed/high) `vsens` - As of Claude Code v2.1.210, the auto-mode permission classifier defaults to Claude Sonnet 5 for external sessions rather than the session's `/model` selection; the default is validated on the session's first au ...

  *Applies:* Claude Code auto-mode classifier model; default changed at v2.1.210 (2026-07-14). Claude Code documentation — Choose a permission mode

- `claude-hooks-permissions-22` (confirmed/high) `vsens` - The Anthropic engineering deep-dive 'claude-code-auto-mode' (published Mar 25, 2026; no updated date shown) states the transcript classifier runs on Sonnet 4.6 and reports full-pipeline measured rates of 0.4% f ...

  *Applies:* Anthropic engineering post as of 2026-07-30; rates characterize the Sonnet 4.6-era auto-mode classifier. Anthropic Engineering — Claude Code auto mode classifier deep dive

- `claude-hooks-permissions-23` (confirmed/high) `vsens` - As of 2026-07-30 the Claude Code changelog is dated and its latest entry is v2.1.220 (July 25, 2026), with v2.1.210 dated July 14 and v2.1.211 dated July 15, 2026. No changelog entry through v2.1.220 reverses t ...

  *Applies:* Claude Code changelog state on 2026-07-30 (latest v2.1.220). Claude Code documentation — Changelog

- `claude-hooks-permissions-24` (confirmed/high) `vsens` - As of Claude Code v2.1.218 (July 22, 2026), when auto mode is available the `useAutoModeDuringPlan` setting is on by default, so during plan mode the classifier reviews shell commands (and protected-path writes ...

  *Applies:* Claude Code auto-mode/plan-mode behavior new at v2.1.218 (2026-07-22); in-window drift. Claude Code documentation — Choose a permission mode

### claude-context-memory-skills (30)

- `claude-context-memory-skills-01` (confirmed/high) `vsens` - On the Anthropic API, Claude Code's Sonnet 5 always runs with the 1M-token context window; there is no 200K variant and no [1m] suffix to select for it.

*Applies:* Claude Code v2.1.197+ on the Anthropic API (Sonnet 5) Model configuration — Sonnet 5 context window

- `claude-context-memory-skills-02` (confirmed/high) `vsens` - On the Anthropic API a Sonnet 5 session auto-compacts before the window fills, at about 967K tokens by default; the CLAUDE_CODE_AUTO_COMPACT_WINDOW environment variable changes that threshold.

  *Applies:* Claude Code v2.1.197+ on the Anthropic API (Sonnet 5) Model configuration — Sonnet 5 context window

- `claude-context-memory-skills-03` (confirmed/high) `vsens` - The sonnet[1m] alias has no effect when 'sonnet' already resolves to Sonnet 5 with its native 1M window, but behind an LLM gateway (ANTHROPIC_BASE_URL) it selects the 1M window for Sonnet 5 because Claude Code ...

  *Applies:* Claude Code v2.1.197+ (Sonnet 5); LLM-gateway deployments Model configuration — Model aliases / Sonnet 5 context window

- `claude-context-memory-skills-04` (confirmed/high) `vsens` - Fable 5, Sonnet 5, Opus 4.6 and later, and Sonnet 4.6 support a 1M-token context window; on the Anthropic API, Fable 5, Sonnet 5, and Opus 4.7 and later always run with the 1M window.

  *Applies:* Claude Code v2.1.x (Anthropic API and per-plan availability) Model configuration — Extended context

- `claude-context-memory-skills-05` (confirmed/high) `vsens` - Setting CLAUDE_CODE_DISABLE_1M_CONTEXT=1 removes 1M model variants from the model picker and treats Sonnet 5 sessions as having a 200K window (which then auto-compacts at the 200K boundary).

  *Applies:* Claude Code v2.1.x Model configuration — Extended context / Sonnet 5 context window

- `claude-context-memory-skills-06` (confirmed/high) `vsens` - CLAUDE_AUTOCOMPACT_PCT_OVERRIDE sets the percentage (1-100) of the auto-compaction window at which auto-compaction triggers; it can only lower the threshold (values above the default have no effect) and applies ...

  *Applies:* Claude Code v2.1.x Environment variables — CLAUDE_AUTOCOMPACT_PCT_OVERRIDE

- `claude-context-memory-skills-07` (confirmed/high) `vsens` - CLAUDE_CODE_AUTO_COMPACT_WINDOW sets the context capacity in tokens used for auto-compaction calculations; it defaults to the model's context window (200K for standard models, 1M for extended-context models) ex ...

  *Applies:* Claude Code v2.1.x Environment variables — CLAUDE_CODE_AUTO_COMPACT_WINDOW

- `claude-context-memory-skills-08` (confirmed/high) `vsens` - For standard 200K models such as Sonnet 4.6 and Opus 4.6 without extended context, Claude Code auto-compacts at the 200K boundary by default; the docs describe this boundary qualitatively and publish no numeric ...

  *Applies:* Claude Code v2.1.x (standard 200K-window models) Environment variables — CLAUDE_AUTOCOMPACT_PCT_OVERRIDE

- `claude-context-memory-skills-09` (confirmed/high) `vsens` - Claude Code loads the first 200 lines of MEMORY.md, or the first 25KB, whichever comes first, into context at the start of every conversation; content beyond that threshold is not loaded at session start.

  *Applies:* Claude Code v2.1.x (auto memory) How Claude remembers your project — How it works

- `claude-context-memory-skills-10` (confirmed/high) `vsens` - As of Claude Code v2.1.211, the MEMORY.md over-limit check measures only the content that loads: YAML frontmatter and block-level HTML comments are stripped before the index is loaded and do not count toward th ...

  *Applies:* Claude Code v2.1.211+ (auto memory) How Claude remembers your project — How it works

- `claude-context-memory-skills-11` (confirmed/high) `vsens` - As of Claude Code v2.1.210, after Claude writes to MEMORY.md, Claude Code measures the file against the 200-line and 25KB read limits, reminds Claude to shorten it when near a limit, and returns an error tellin ...

  *Applies:* Claude Code v2.1.210+ (auto memory) How Claude remembers your project — How it works

- `claude-context-memory-skills-12` (confirmed/high) `vsens` - Anthropic's docs recommend targeting under 200 lines per CLAUDE.md file, stating longer files 'consume more context and reduce adherence', and note CLAUDE.md files are loaded in full regardless of length; no me ...

  *Applies:* Claude Code v2.1.x docs guidance (vendor assertion) How Claude remembers your project — Write effective instructions

- `claude-context-memory-skills-13` (confirmed/high) `vsens` - As of Claude Code v2.1.198, subagents run in the background by default; Claude runs a subagent in the foreground only when it needs the result before continuing, and background subagents run with a narrower bui ...

  *Applies:* Claude Code v2.1.198+ (subagents) Create custom subagents — Run subagents in foreground or background / Available tools

- `claude-context-memory-skills-14` (confirmed/high) `vsens` - The sub-agents docs attribute to v2.1.186 the behavior where a background subagent's permission prompt surfaces in the main session naming the asking subagent (approve to continue, Esc to deny that one tool cal ...

  *Applies:* Claude Code v2.1.186+ (background subagents) Create custom subagents — Run subagents in foreground or background

- `claude-context-memory-skills-15` (confirmed/high) `vsens` - Claude Code subagent scope precedence, highest to lowest, is: 1 managed settings, 2 the --agents CLI flag (current session), 3 project .claude/agents/, 4 user ~/.claude/agents/, 5 a plugin's agents/ directory; ...

  *Applies:* Claude Code v2.1.x (subagents) Create custom subagents — Choose the subagent scope

- `claude-context-memory-skills-16` (confirmed/high) `vsens` - By default Claude can spawn at most 200 subagents per session; CLAUDE_CODE_MAX_SUBAGENTS_PER_SESSION raises it to any positive whole number (no upper bound, cannot be turned off), and it requires Claude Code v2 ...

*Applies:* Claude Code v2.1.212+ (session subagent limit) Create custom subagents — Session subagent limit

- `claude-context-memory-skills-17` (confirmed/high) `vsens` - By default, when 20 subagents are running concurrently in a session, spawning another with the Agent tool fails with 'Concurrent subagent limit reached'; CLAUDE_CODE_MAX_CONCURRENT_SUBAGENTS changes the limit, ...

  *Applies:* Claude Code v2.1.217+ (concurrent subagent limit) Create custom subagents — Concurrent subagent limit

- `claude-context-memory-skills-18` (confirmed/high) `vsens` - As of Claude Code v2.1.219 the default subagent nesting depth is three layers below the main conversation, set via CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH (introduced v2.1.217); v2.1.172 through v2.1.216 nested up ...

  *Applies:* Claude Code v2.1.217+ (subagent nesting depth), default 3 as of v2.1.219 Create custom subagents — Let subagents spawn their own subagents

- `claude-context-memory-skills-19` (confirmed/high) `vsens` - Agent team teammates and workflow agents do NOT count toward the subagent session or concurrent limits; they follow their own separate limits.

  *Applies:* Claude Code v2.1.x (subagent limits vs agent teams/workflows) Create custom subagents — Concurrent subagent limit / Session subagent limit

- `claude-context-memory-skills-20` (confirmed/high) `vsens` - A subagent can be given an isolated copy of the repository by setting isolation: worktree in its frontmatter, which runs it in a temporary git worktree branched by default from the default branch; the worktree ...

  *Applies:* Claude Code v2.1.x (subagents; worktree checks tightened in v2.1.203/2.1.210/2.1.216) Create custom subagents — Supported frontmatter fields (isolation)

- `claude-context-memory-skills-21` (confirmed/high) `vsens` - In the skill listing, each skill's combined 'description' plus 'when_to_use' text is truncated at 1,536 characters to reduce context usage; the cap is configurable via the skillListingMaxDescChars setting.

  *Applies:* Claude Code v2.1.x (skills) Extend Claude with skills — Frontmatter reference / Skill descriptions are cut short

- `claude-context-memory-skills-22` (confirmed/high) `vsens` - The skill-listing character budget scales at 1% of the model's context window by default; it is configurable via skillListingBudgetFraction or a fixed count via the SLASH_COMMAND_TOOL_CHAR_BUDGET environment va ...

  *Applies:* Claude Code v2.1.x (skills listing) Extend Claude with skills — Skill descriptions are cut short

- `claude-context-memory-skills-23` (confirmed/high) `vsens` - After auto-compaction, Claude Code re-attaches the most recent invocation of each invoked skill, keeping the first 5,000 tokens of each within a combined 25,000-token budget filled from the most-recently-invoke ...

  *Applies:* Claude Code v2.1.x (skills + compaction) Extend Claude with skills — Skill content lifecycle

- `claude-context-memory-skills-24` (confirmed/high) `vsens` - By default MCP tool schemas stay deferred (only tool names load); ENABLE_TOOL_SEARCH=auto loads schemas upfront when they fit within 10% of the context window, and ENABLE_TOOL_SEARCH=false loads every schema.

  *Applies:* Claude Code v2.1.x (MCP tool search) Explore the context window — MCP tools (deferred)

- `claude-context-memory-skills-25` (confirmed/high) `vsens` - Anthropic's cost docs state agent teams use approximately 7x more tokens than standard sessions specifically when teammates run in plan mode, because each teammate maintains its own context window as a separate ...

  *Applies:* Claude Code v2.1.x (agent teams, plan mode) Manage costs effectively — Manage agent team costs

- `claude-context-memory-skills-26` (confirmed/high) `vsens` - Agent teams are experimental and disabled by default; they are enabled by setting CLAUDE_CODE_EXPERIMENTAL_AGENT_TEAMS=1 in settings.json or the environment, without which no team is set up and Claude does not ...

  *Applies:* Claude Code v2.1.178+ (agent teams; page documents behavior as of v2.1.178) Orchestrate teams of Claude Code sessions — Enable agent teams / Limitations

- `claude-context-memory-skills-27` (confirmed/high) `vsens` - Anthropic's agent-teams docs recommend starting with 3-5 teammates for most workflows and about 5-6 tasks per teammate; both figures are presented as heuristics ('works well', 'keeps everyone productive') with ...

  *Applies:* Claude Code v2.1.178+ (agent teams guidance) Orchestrate teams of Claude Code sessions — Choose an appropriate team size / Size tasks appropriately

- `claude-context-memory-skills-28` (confirmed/high) `vsens` - The agent-teams path provides no worktree or file-isolation mechanism for teammates; the only documented mitigation for two teammates editing the same file (which 'leads to overwrites') is prompt-level file-own ...

  *Applies:* Claude Code v2.1.178+ (agent teams) Orchestrate teams of Claude Code sessions — Avoid file conflicts

- `claude-context-memory-skills-29` (confirmed/high) - Anthropic's 'Effective context engineering for AI agents' article (dated September 29, 2025) contains no quantitative context-management performance figures (no 29%/39%/84% numbers), names no eval or baseline, ...

  *Applies:* Anthropic Engineering blog (2025-09-29), vendor-agnostic conceptual guidance Effective context engineering for AI agents

- `claude-context-memory-skills-30` (confirmed/high) `vsens` - In Claude Code, Sonnet 5 requires v2.1.197 or later and Opus 5 requires v2.1.219 or later; on the Anthropic API the 'sonnet' alias resolves to Sonnet 5 and the 'opus' alias resolves to Opus 5 (before v2.1.219 ' ...

  *Applies:* Claude Code v2.1.197+ (Sonnet 5), v2.1.219+ (Opus 5), Anthropic API alias resolution Model configuration — Model aliases / Available models

### codex-config-hooks (23)

- `codex-config-hooks-01` (confirmed/high) `vsens` - The current stable Codex CLI release is 0.146.0, published 2026-07-29 (git tag rust-v0.146.0), installable via `npm install -g @openai/codex@0.146.0`.

  *Applies:* Codex CLI stable channel as of 2026-07-30 (supersedes prior baseline stable 0.144.6) Codex Changelog

- `codex-config-hooks-02` (confirmed/high) `vsens` - The latest Codex CLI pre-release is 0.147.0-alpha.2 (git tag rust-v0.147.0-alpha.2, prerelease=true), published 2026-07-30T01:04:28Z; the alpha line has advanced from 0.145.0-alpha to the 0.147.0-alpha series.

  *Applies:* Codex CLI alpha/pre-release channel as of 2026-07-30 openai/codex releases API

- `codex-config-hooks-03` (confirmed/high) `vsens` - The Codex 0.146.0 (2026-07-29) release notes contain no entries touching hooks, config.toml, AGENTS.md, trust, or requirements.toml; its changes are Agent Plugins manifests/marketplaces, thread forking with pag ...

  *Applies:* Codex CLI 0.146.0 release notes Codex rust-v0.146.0 release notes

- `codex-config-hooks-04` (confirmed/high) `vsens` - Codex 0.145.0 (released 2026-07-21) shipped a hook-related bug fix: correctly quoting hook commands on Windows, delivered alongside native exec-server sandboxing and network-proxy enforcement work.

  *Applies:* Codex CLI 0.145.0 release notes (Windows command hooks) Codex rust-v0.145.0 release notes

- `codex-config-hooks-05` (confirmed/high) `vsens` - The Codex hooks documentation enumerates 11 hook events: SessionStart, SessionEnd, PreToolUse, PostToolUse, PermissionRequest, PreCompact, PostCompact, UserPromptSubmit, SubagentStart, SubagentStop, and Stop.

  *Applies:* Codex CLI hooks docs as of 2026-07-30 (docs track main; may lead the 0.146.0 release) Codex Hooks documentation

- `codex-config-hooks-06` (confirmed/high) `vsens` - Only hook handlers of `type: "command"` execute in Codex today; `prompt` and `agent` handler types are parsed but skipped.

  *Applies:* Codex CLI hooks docs / config.toml reference as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-07` (confirmed/high) `vsens` - Hook matcher semantics vary by event: PreToolUse/PostToolUse/PermissionRequest match on tool name; PreCompact/PostCompact match on the compaction trigger (`manual` or `auto`); SessionStart matches on session so ...

  *Applies:* Codex CLI hooks docs as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-08` (confirmed/high) `vsens` - A PreToolUse hook can block a tool via `hookSpecificOutput` with `{hookEventName:"PreToolUse", permissionDecision:"deny", permissionDecisionReason}` (or the legacy `{decision:"block", reason}` form), and can re ...

  *Applies:* Codex CLI hooks docs as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-09` (confirmed/high) `vsens` - A PermissionRequest hook returns its decision via `hookSpecificOutput` with `{hookEventName:"PermissionRequest", decision:{behavior:"allow"|"deny", message}}`.

  *Applies:* Codex CLI hooks docs as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-10` (confirmed/high) `vsens` - The common Codex hook output fields (supported by most events) are `continue`, `stopReason`, `systemMessage`, and `suppressOutput`.

  *Applies:* Codex CLI hooks docs as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-11` (confirmed/high) `vsens` - Codex records hook trust against the hook's current hash and requires the user to review and trust the exact hook definition before a non-managed command hook runs; new or changed hooks are marked for review an ...

  *Applies:* Codex CLI hooks trust model as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-12` (confirmed/high) `vsens` - The `--dangerously-bypass-hook-trust` CLI flag runs hooks without establishing persistent hash-level trust, intended for one-off execution of pre-vetted hook sources.

  *Applies:* Codex CLI hooks docs as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-13` (confirmed/high) `vsens` - Managed hooks originating from system, MDM, cloud, or `requirements.toml` sources bypass the individual per-hook trust review that non-managed command hooks require.

  *Applies:* Codex CLI hooks trust model as of 2026-07-30 Codex Hooks documentation

- `codex-config-hooks-14` (confirmed/high) `vsens` - The `projects.<path>.trust_level` config key marks a project as `"trusted"` or `"untrusted"`; untrusted projects skip project-scoped `.codex/` configuration layers.

  *Applies:* Codex CLI config.toml reference as of 2026-07-30 Codex config.toml Configuration Reference

- `codex-config-hooks-15` (confirmed/high) `vsens` - Project-local hooks load only when the project's `.codex/` layer is trusted; combined with per-hook hash trust, non-managed project hooks face two gates (project-layer trust to load, plus hash trust to execute) ...

  *Applies:* Codex CLI advanced-config + hooks docs as of 2026-07-30 Codex Advanced Configuration

- `codex-config-hooks-16` (confirmed/high) `vsens` - Codex supports lifecycle hooks configured inline in config.toml via `[hooks]` tables using the same event schema as hooks.json, gated by the `features.hooks` boolean, which the config reference labels stable an ...

  *Applies:* Codex CLI config.toml reference as of 2026-07-30 Codex config.toml Configuration Reference

- `codex-config-hooks-17` (confirmed/high) `vsens` - The config key `hooks.<Event>[].hooks[].additionalContextLimit` (integer) sets an approximate per-handler token threshold for spilling oversized `additionalContext` to disk and showing the model a shorter previ ...

*Applies:* Codex CLI config.toml reference as of 2026-07-30 Codex config.toml Configuration Reference

- `codex-config-hooks-18` (confirmed/high) `vsens` - The config key `hooks.<Event>[].hooks[].commandWindows` (string) provides a Windows-only command override for command hooks.

  *Applies:* Codex CLI config.toml reference as of 2026-07-30 Codex config.toml Configuration Reference

- `codex-config-hooks-19` (confirmed/high) `vsens` - The Codex AGENTS.md documentation describes `project_doc_max_bytes` as a combined/accumulated cap: Codex skips empty files and stops adding files once the combined size reaches the limit, with a default of 32 K ...

  *Applies:* Codex CLI AGENTS.md docs as of 2026-07-30 Codex AGENTS.md documentation

- `codex-config-hooks-20` (confirmed/high) `vsens` - The Codex advanced-configuration page describes `project_doc_max_bytes` as a per-file cap, wording it as "how much to read from each AGENTS.md file" — directly conflicting with the AGENTS.md page's combined-siz ...

  *Applies:* Codex CLI advanced-config docs vs AGENTS.md docs as of 2026-07-30 Codex Advanced Configuration

- `codex-config-hooks-21` (confirmed/high) `vsens` - Codex AGENTS.md discovery concatenates instruction files from the Git root downward with later (deeper) files overriding earlier ones, checking `AGENTS.override.md` then `AGENTS.md` then configured fallback fil ...

  *Applies:* Codex CLI AGENTS.md docs as of 2026-07-30 Codex AGENTS.md documentation

- `codex-config-hooks-22` (confirmed/high) `vsens` - On the openai/codex `main` branch as of 2026-07-30, the `CoreToolRuntime` trait in codex-rs/core/src/tools/registry.rs defines `pre_tool_use_payload` with a default implementation that returns `Some(PreToolUseP ...

  *Applies:* openai/codex main branch source as of 2026-07-30 (NOT confirmed present in the 0.146.0 stable release; docs wa ... openai/codex core/src/tools/registry.rs (main)

- `codex-config-hooks-23` (confirmed/high) `vsens` - GitHub issue openai/codex#20204 (opened 2026-04-29, still open as of 2026-07-30) reports that PreToolUse/PostToolUse/PermissionRequest hooks emit only for the shell, unified_exec, apply_patch, and mcp handlers, ...

  *Applies:* Codex CLI hook tool-coverage as reported against the 0.14x line (issue open, unresolved in tracker) openai/codex issue #20204

### codex-exec-sandbox (24)

- `codex-exec-sandbox-01` (confirmed/high) `vsens` - In Codex CLI, `codex exec` streams progress to stderr and prints only the final agent message to stdout; `-o` / `--output-last-message <path>` additionally writes that final message to a file WHILE STILL printing ...

  *Applies:* Codex CLI non-interactive mode (learn.chatgpt.com docs current 2026-07-30; latest stable rust-v0.146.0) Non-interactive mode | ChatGPT Learn

- `codex-exec-sandbox-02` (confirmed/high) `vsens` - `codex exec --json` turns stdout into a JSON Lines (JSONL) stream; the documented event types are thread.started, turn.started, turn.completed, turn.failed, item.*, and error.

  *Applies:* Codex CLI non-interactive mode (docs current 2026-07-30) Non-interactive mode | ChatGPT Learn

- `codex-exec-sandbox-03` (confirmed/high) `vsens` - The `turn.completed` event's `usage` object contains input_tokens, cached_input_tokens, output_tokens, and reasoning_output_tokens.

  *Applies:* Codex CLI non-interactive JSONL output (docs current 2026-07-30) Non-interactive mode | ChatGPT Learn

- `codex-exec-sandbox-04` (confirmed/high) `vsens` - `codex exec --full-auto` is deprecated; the non-interactive and approvals docs direct users to `--sandbox workspace-write` instead.

  *Applies:* Codex CLI non-interactive mode (docs current 2026-07-30) Non-interactive mode | ChatGPT Learn

- `codex-exec-sandbox-05` (confirmed/high) `vsens` - The non-interactive-mode docs state that `codex exec` runs in the read-only sandbox by default; the CLI command reference separately describes `--sandbox` as otherwise inheriting from configuration ('defaults t ...

  *Applies:* Codex CLI non-interactive mode (docs current 2026-07-30) Non-interactive mode | ChatGPT Learn

- `codex-exec-sandbox-06` (confirmed/high) `vsens` - Codex has exactly three sandbox modes: read-only, workspace-write, and danger-full-access; workspace-write is documented as the default low-friction mode for local (interactive) work.

  *Applies:* Codex CLI sandboxing (docs current 2026-07-30) Sandbox | ChatGPT Learn

- `codex-exec-sandbox-07` (confirmed/high) `vsens` - The `--ask-for-approval` flag's documented values are untrusted, on-request, and never; `on-failure` does NOT appear in the current CLI command reference or the approval_policy enum.

  *Applies:* Codex CLI (developer-commands reference current 2026-07-30) Developer commands (CLI reference) | ChatGPT Learn

- `codex-exec-sandbox-08` (confirmed/high) `vsens` - The `approval_policy` config key accepts untrusted, on-request, never, or a granular table form `{ granular = { sandbox_approval, rules, mcp_elicitations, request_permissions, skill_approval } }`; the config re ...

  *Applies:* Codex CLI config.toml (config-reference current 2026-07-30) Configuration Reference | ChatGPT Learn

- `codex-exec-sandbox-09` (confirmed/high) `vsens` - Codex non-interactive docs instruct setting CODEX_API_KEY only for the single `codex exec` invocation, scoping the credential to that process rather than exporting it into the surrounding shell/CI environment.

  *Applies:* Codex CLI non-interactive mode (docs current 2026-07-30) Non-interactive mode | ChatGPT Learn

- `codex-exec-sandbox-10` (confirmed/high) `vsens` - The openai/codex-action GitHub Action authenticates via the `openai-api-key` input routed through an internal proxy (exposed to the CLI as PROXY_API_KEY) and does NOT use a CODEX_API_KEY variable; it reduces pr ...

  *Applies:* openai/codex-action, main branch action.yml (accessed 2026-07-30) [openai/codex-action action.yml (raw)]

- `codex-exec-sandbox-11` (confirmed/high) `vsens` - openai/codex-action gates who may trigger it via `allow-users` (comma-separated GitHub usernames), `allow-bots` (default false), and `allow-bot-users`, alongside a repository write-access check.

  *Applies:* openai/codex-action, main branch action.yml (accessed 2026-07-30) [openai/codex-action action.yml (raw)]

- `codex-exec-sandbox-12` (confirmed/high) `vsens` - openai/codex-action exposes a legacy `sandbox` input (workspace-write | read-only | danger-full-access) and a `permission-profile` input (selected via `default_permissions`); the action README states permission ...

  *Applies:* openai/codex-action, main branch (accessed 2026-07-30); permission-profile needs Codex CLI >=0.138.0 [openai/codex-action action.yml (raw)]

- `codex-exec-sandbox-13` (confirmed/high) `vsens` - Codex CLI release rust-v0.144.6 (2026-07-18) refreshed the bundled instructions for GPT-5.6 Sol, Terra, and Luna and corrected their context windows to 272,000 tokens (PRs #33972, #34009).

  *Applies:* Codex CLI rust-v0.144.6 (2026-07-18); models GPT-5.6 Sol/Terra/Luna as driven by Codex [Release 0.144.6 · openai/codex]

- `codex-exec-sandbox-14` (confirmed/high) `vsens` - Session archiving shipped in Codex CLI rust-v0.136.0 (2026-06-01): sessions can be archived from the TUI with `/archive` or from the CLI with `codex archive` / `codex unarchive`, and archived sessions are prote ...

  *Applies:* Codex CLI rust-v0.136.0 (2026-06-01) and later [Release 0.136.0 · openai/codex]

- `codex-exec-sandbox-15` (confirmed/high) `vsens` - `codex fork`, `codex archive`, and `codex unarchive` are documented Codex CLI subcommands in the developer-commands (CLI) reference, even though the higher-level Codex CLI overview page does not list them.

  *Applies:* Codex CLI (developer-commands reference current 2026-07-30) [Developer commands (CLI reference) | ChatGPT Learn]

- `codex-exec-sandbox-16` (confirmed/high) `vsens` - The Codex linux-sandbox now uses Bubblewrap (bwrap) plus a seccomp network filter as the default enforcement on Linux, with Landlock+mount protection retained only as a legacy fallback (selectable via `features ...

  *Applies:* Codex CLI codex-rs linux-sandbox crate, main branch (accessed 2026-07-30) [codex-rs/linux-sandbox/README.md (raw)]

- `codex-exec-sandbox-17` (confirmed/high) `vsens` - Codex's initial skills list is capped at 'at most 2% of the model's context window, or 8,000 characters when the context window is unknown'; the budget applies only to the initial list, and the full SKILL.md is ...

*Applies:* Codex skills / progressive disclosure (build-skills docs current 2026-07-30) Build skills | ChatGPT Learn

- `codex-exec-sandbox-18` (confirmed/high) `vsens` - Codex skills support explicit invocation (run `/skills` or type `$` in Codex CLI / IDE extension; `@` in ChatGPT) and implicit invocation (Codex selects a skill when the task matches the skill's `description`).

  *Applies:* Codex skills (build-skills docs current 2026-07-30) Build skills | ChatGPT Learn

- `codex-exec-sandbox-19` (confirmed/high) `vsens` - Codex ships three built-in subagents: `default` (general-purpose fallback), `worker` (execution-focused for implementation and fixes), and `explorer` (read-heavy codebase exploration).

  *Applies:* Codex subagents (agent-configuration/subagents docs current 2026-07-30) Subagents | ChatGPT Learn

- `codex-exec-sandbox-20` (confirmed/high) `vsens` - Custom Codex subagents are standalone TOML files in `~/.codex/agents/` (personal) or `.codex/agents/` (project-scoped) and can override settings including sandbox_mode, model, model_reasoning_effort, mcp_server ...

  *Applies:* Codex subagents (agent-configuration/subagents docs current 2026-07-30) Subagents | ChatGPT Learn

- `codex-exec-sandbox-21` (confirmed/high) `vsens` - `agents.max_concurrent_threads_per_session` is a real Codex config key (caps concurrently-open spawned-agent threads, excluding the primary; auto-defaulted when unset); `features.multi_agent_v2` and `agents.job ...

  *Applies:* Codex CLI config.toml (config-reference + subagents docs current 2026-07-30) Configuration Reference | ChatGPT Learn

- `codex-exec-sandbox-22` (confirmed/high) `vsens` - The MCP `default_tools_approval_mode` config key accepts auto, prompt, writes, or approve (writes = prompt only for tools not marked read-only); a per-tool override exists via `tools.<tool>.approval_mode`. The ...

  *Applies:* Codex CLI mcp_servers config (config-reference + MCP docs current 2026-07-30) Model Context Protocol | ChatGPT Learn

- `codex-exec-sandbox-23` (confirmed/high) `vsens` - Codex CLI rust-v0.146.0 (2026-07-29) detaches non-interactive subprocesses from stdin (#34612).

  *Applies:* Codex CLI rust-v0.146.0 (2026-07-29) Release 0.146.0 · openai/codex

- `codex-exec-sandbox-24` (confirmed/high) `vsens` - As of 2026-07-30 the latest stable Codex CLI release is rust-v0.146.0 (2026-07-29), which supersedes rust-v0.144.6; rust-v0.147.0 is in alpha.

  *Applies:* Codex CLI GitHub releases index (accessed 2026-07-30) Releases · openai/codex

### security-sandboxing-injection (9)

- `security-sandboxing-injection-17` (confirmed/high) `vsens` - Claude Code added the `sandbox.filesystem.disabled` setting in v2.1.216 (released 2026-07-20). When set to `true`, the sandbox skips filesystem isolation while keeping network isolation, so sandboxed commands g ...

*Applies:* Claude Code v2.1.216+ CLI, sandbox enabled (macOS/Linux/WSL2) Configure the sandboxed Bash tool - Claude Code docs

- `security-sandboxing-injection-18` (confirmed/high) `vsens` - Claude Code added the `sandbox.network.strictAllowlist` setting in v2.1.219 (released 2026-07-24). When set to `true`, the sandbox denies sandboxed commands access to any host outside the allowlist instead of p ...

  *Applies:* Claude Code v2.1.219+ CLI, sandbox enabled Configure the sandboxed Bash tool - Claude Code docs

- `security-sandboxing-injection-19` (confirmed/high) `vsens` - Even with `sandbox.network.strictAllowlist` (v2.1.219+) enabled, Claude Code's built-in sandbox proxy still makes its allow/deny decision from the client-supplied hostname and by default does not terminate or i ...

  *Applies:* Claude Code v2.1.x CLI (strictAllowlist v2.1.219+, tlsTerminate v2.1.199+); default non-TLS-inspecting built-i ... Configure the sandboxed Bash tool - Claude Code docs (Network isolation, Security limitations)

- `security-sandboxing-injection-20` (confirmed/high) - The OWASP MCP Security Cheat Sheet (accessed 2026-07-30), Best Practices section 12 'Prompt Injection via Tool Return Values,' recommends treating every tool return value as untrusted data rather than instructi ...

  *Applies:* Vendor-agnostic MCP client/server design; OWASP MCP Security Cheat Sheet OWASP MCP Security Cheat Sheet

- `security-sandboxing-injection-21` (corrected/high) - The OWASP MCP Security Cheat Sheet (accessed 2026-07-30) does recommend every control the claim lists, but they are split across two numbered Best Practices subsections rather than all residing in section 2. Se ...

  *Applies:* Vendor-agnostic MCP tooling; OWASP MCP Security Cheat Sheet OWASP MCP Security Cheat Sheet

- `security-sandboxing-injection-22` (corrected/high) - OWASP MCP Security Cheat Sheet (accessed 2026-07-30) Best Practices section 3 'Sandbox and Isolate MCP Servers' recommends running local MCP servers in sandboxed environments (containers, chroot, application sa ...

  *Applies:* Vendor-agnostic MCP deployment; OWASP MCP Security Cheat Sheet OWASP MCP Security Cheat Sheet

- `security-sandboxing-injection-23` (confirmed/high) - The OWASP MCP Security Cheat Sheet (accessed 2026-07-30) recommends, for MCP authentication and least privilege: OAuth 2.0 with PKCE for remote-server authorization flows; scoped per-server credentials that are ...

  *Applies:* Vendor-agnostic MCP authentication/authorization; OWASP MCP Security Cheat Sheet OWASP MCP Security Cheat Sheet

- `security-sandboxing-injection-24` (confirmed/high) `vsens` - NIST Special Publication 800-218A, 'Secure Software Development Practices for Generative AI and Dual-Use Foundation Models:

An SSDF Community Profile' (authors Harold Booth, Murugiah Souppaya, Apostol Vassilev, ...

*Applies:* NIST SP 800-218A (final, 2024-07-26); governance / secure-development baseline NIST SP 800-218A (Final) publication page

- `security-sandboxing-injection-25` (confirmed/high) `vsens` - In Claude Code's auto-allow sandbox mode, `rm` / `rmdir` commands targeting `/`, the user's home directory, or other critical system paths still trigger a permission prompt (or an auto-mode classifier check), wit ...

  *Applies:* Claude Code v2.1.x CLI; destructive-command classifier routing and plan-mode auto-mode require v2.1.218+ Configure the sandboxed Bash tool - Claude Code docs (Sandbox modes)

### evals-judging (14)

- `evals-judging-20` (confirmed/high) `vsens` - As of 2026-07-30, swebench.com presents the SWE-bench family as four execution-based issue-resolution variants (SWE-bench, SWE-bench Verified, SWE-bench Multimodal, SWE-bench Multilingual, plus SWE-bench Lite) ...

  *Applies:* swebench.com leaderboard site, snapshot 2026-07-30; SWE-bench benchmark family (vendor-agnostic harnesses) SWE-bench leaderboards (official site)

- `evals-judging-25` (confirmed/high) `vsens` - METR's early-2025 randomized controlled trial found AI tools made 16 experienced open-source developers 19% SLOWER on 246 real repository tasks (CI +2% to +39%), even though the same developers forecast a 24% s ...

  *Applies:* METR RCT, published 2025-07-10; Cursor Pro + Claude 3.5/3.7 Sonnet era Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity (METR)

- `evals-judging-26` (confirmed/high) `vsens` - METR's 2026-02-24 update reports that its follow-up uplift experiment (57 developers, 800+ tasks) was undermined by selection bias (developers declined to participate rather than work without AI; a pay cut from ...

  *Applies:* METR uplift-experiment update, published 2026-02-24 (supersedes the 2025-07-10 study) We are Changing our Developer Productivity Experiment Design (METR)

- `evals-judging-27` (confirmed/high) `vsens` - RuVerBench (arXiv:2606.29920, THU-KEG) is a benchmark of 2,458 human-labeled rubric-verification instances split into Deep Research (1,615) and Agentic Coding (843); each instance pairs a model output with a ru ...

  *Applies:* RuVerBench, arXiv:2606.29920 (June 2026); LLM-as-a-judge rubric verification for agentic coding + deep researc ... Can LLM-as-a-Judge Reliably Verify Rubrics in Agentic Scenarios? (RuVerBench)

- `evals-judging-28` (confirmed/high) `vsens` - On RuVerBench's Agentic Coding split (18 judge models), the top verifiers by Avg BAcc are GPT-5.4 (89.4), Gemini-3.1 Pro Preview (86.5), Claude Opus 4.7 (85.0), Kimi K2.6 (84.3, best open-weight), and DeepSeek ...

  *Applies:* RuVerBench Table 2, arXiv:2606.29920 (June 2026); named 2026 frontier judge models RuVerBench main leaderboard (arXiv:2606.29920, Table 2)

- `evals-judging-29` (confirmed/high) `vsens` - RuVerBench finds majority voting (self-consistency) over a judge yields effective but diminishing returns: gains are mostly reached within three to five votes and the curves flatten afterward (larger benefit in ...

  *Applies:* RuVerBench judging-strategy analysis (fixed 20% subset, 6 models), arXiv:2606.29920 (June 2026) RuVerBench majority-voting analysis (arXiv:2606.29920, sec. 4.4)

- `evals-judging-30` (confirmed/high) `vsens` - RuVerBench shows top judges are not interchangeable: among Gemini-3.1 Pro Preview, GPT-5.4, and Claude Opus 4.7 the average error-set overlap is only 16.1% (Deep Research) and 20.6% (Agentic Coding), and models ...

  *Applies:* RuVerBench error-profile analysis, arXiv:2606.29920 (June 2026) RuVerBench error-profile analysis (arXiv:2606.29920, sec. 4.3)

- `evals-judging-31` (confirmed/high) `vsens` - RuVerBench finds a batching accuracy cost: once a single judge call bundles four or five rubrics, every tested model loses accuracy on Agentic Coding, often by double digits (Deep Research shows only small/mixe ...

  *Applies:* RuVerBench batching analysis, arXiv:2606.29920 (June 2026) RuVerBench batching analysis (arXiv:2606.29920, sec. 4.4)

- `evals-judging-32` (confirmed/high) `vsens` - LongJudgeBench (arXiv:2606.01629, Tsinghua/USTB) evaluates LLM judges on outputs averaging 9,249.7 tokens (versus MT-Bench 194 and RewardBench 2 312.3), and finds long-form judging largely unsolved: across 8 ju ...

  *Applies:* LongJudgeBench, arXiv:2606.01629 (submitted 2026-06-01); judges incl. GPT-5.2, DeepSeek-V4-Flash, GLM-5.1, Kim ... Benchmarking LLM-as-a-Judge for Long-Form Output Evaluation (LongJudgeBench)

- `evals-judging-33` (confirmed/high) `vsens` - CodeJudgeBench (arXiv:2507.10535) benchmarks 26 LLM-as-a-judge models across code generation, code repair, and unit-test generation, and finds pairwise comparison outperforms scalar pointwise scoring, that simp ...

  *Applies:* CodeJudgeBench, arXiv:2507.10535 (v1 2025-07-14, v2 2025-08-14); LLM-as-judge for coding tasks CodeJudgeBench: Benchmarking LLM-as-a-Judge for Coding Tasks

- `evals-judging-34` (confirmed/high) `vsens` - An audit of reward hackability (arXiv:2606.16062, S. Rajan, 2026-06-14) finds that on a 49-task SWE-bench Verified sample, 28.5% of tasks have test suites weak enough that a Docker-verified INCORRECT patch pass ...

  *Applies:* arXiv:2606.16062; SWE-bench Verified + R2E-Gym task samples; reward-hacking / test-suite-integrity auditing Auditing Reward Hackability in Code RL Training Environments

- `evals-judging-35` (confirmed/high) `vsens` - SWE-bench Pro (Scale AI) is a contamination-resistant successor to SWE-bench Verified: 1,865 execution-verified tasks across 41 professional repositories in Python/Go/TypeScript/JavaScript, partitioned into pub ...

  *Applies:* SWE-bench Pro public leaderboard (labs.scale.com), snapshot 2026-07-30 SWE-bench Pro Public Leaderboard (Scale)

- `evals-judging-36` (confirmed/high) `vsens` - DeepSWE (arXiv:2607.07946, submitted 2026-07-08) is a contamination-resistant benchmark of 113 original long-horizon tasks across 91 repositories and 5 languages, graded by hand-written verifiers that accept an ...

  *Applies:* DeepSWE, arXiv:2607.07946 (2026-07-08); long-horizon coding-agent evaluation with custom verifiers DeepSWE: Measuring Frontier Coding Agents on Original, Long-Horizon Engineering Tasks

- `evals-judging-37` (confirmed/high) `vsens` - The OpenHands Index (launched 2026-01-29) is a multi-task coding-agent leaderboard that aggregates five execution-based benchmarks — issue resolution (SWE-bench Verified), greenfield (commit0), frontend (a filt ...

  *Applies:* OpenHands Index launch write-up, openhands.dev, 2026-01-29 (index snapshot grows over time) Introducing the OpenHands Index (OpenHands blog)

## Rejected claims

- `evals-judging-21` (unsupported) - CodeScaleBench's frozen published analysis suite is named csb-v1-mixed371 and contains 371 tasks (151 SDLC + 220 organizational); the Sourcegraph launch blog rounds this to '370 tasks (150 SDLC + 220 ...

  *Reason:* Content checks out (GitHub README corroborates csb-v1-mixed371 = 371 = 151 SDLC + 220 org; blog rounds to 370 = 150+220), but this is a version_sensitive claim backed ONLY by tier-3 first-party sources (vendor blog + an in-flux 'living benchmark' repo). Per th ...

- `evals-judging-22` (unsupported) - CodeScaleBench uses two verifier types: SDLC tasks are scored by a patch-based SWE-bench-style verifier against a per-task ground_truth.json, while Org (cross-repo) tasks are scored by an artifact ver ...

  *Reason:* The Sourcegraph blog does describe SDLC = patch-based SWE-bench-style verifier against ground_truth.json and Org = 'artifact' verifier comparing answer.json to a curator solution with an oracle-check fallback, so the content is corroborated. But it is a versio ...

- `evals-judging-23` (unsupported) - CodeScaleBench public results were reported for only one harness+model combination, Claude Code with Haiku 4.5; five further harnesses (Codex, Cursor, Gemini, Copilot, OpenHands) are described as supp ...

  *Reason:* The blog does confirm results were reported only for Claude Code + Haiku 4.5 and that six harnesses (Claude Code, Codex, Cursor, Gemini, Copilot, OpenHands) are supported/planned, so the single-harness scope is corroborated. But this version_sensitive claim re ...

- `evals-judging-24` (unsupported) - In CodeScaleBench's single reported harness (Claude Code + Haiku 4.5), adding Sourcegraph MCP code-navigation raised measured retrieval from File recall 0.127->0.277, Precision@5 0.140->0.478, and F1@ ...

  *Reason:* The blog confirms the exact deltas (File recall 0.127->0.277, Precision@5 0.140->0.478, F1@5 0.099->0.262, Org P@5 0.007->0.471, cost $0.73->$0.51), so the numbers are corroborated. However these are version_sensitive vendor-run measurements from a single tier ...

## Drift detected

82 observations vs the 2026-07-20 baseline.

- [claude-loops-scheduling] Current release advanced from the baseline's ambiguous 2.1.214/2.1.215 to v2.1.220 (changelog top entry; contiguous 2.1.216-2.1.220 are all post-baseline).
- [claude-loops-scheduling] MODEL DRIFT: Opus 5 (claude-opus-5) is now the default Opus model as of v2.1.219 - 1M context, fast mode $10/$50 per Mtok; Opus 4.7 removed from fast mode (/fast now = Opus 5 + Opus 4.8). Any loop/goal cost figures keyed to a prior default Opus need re-basing.
- [claude-loops-scheduling] Subagent nesting depth default toggled since baseline: v2.1.217 disabled default nesting, then v2.1.219 set the default to depth 3 (CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH=1 disables). Net current default = 3.
- [claude-loops-scheduling] Agent SDK budget enforcement became subagent-aware in v2.1.217: subagent spend counts toward max_budget_usd, spawns fail with 'Budget limit reached' at the cap, and background subagents are stopped.
- [claude-loops-scheduling] v2.1.218: /code-review now runs as a background subagent when explicitly invoked.
- [claude-loops-scheduling] v2.1.214: EndConversation tool added - a new model-initiated session-termination path (abusive-user / jailbreak).
- [claude-loops-scheduling] v2.1.216: scheduling now hard-fails when the .claude dir or task file is a symlink (previously wrote through the link).
- [claude-loops-scheduling] Resolved baseline version conflicts against the raw changelog: CLAUDE_CODE_MAX_SUBAGENTS_PER_SESSION => v2.1.212 (not 2.1.199); '/loop hiding session from /resume' fix => v2.1.211 (not 2.1.178).
- [claude-loops-scheduling] Open Q5 is now documented rather than inferred: /verify and /code-review are disable-model-invocation:true skills, so in a /loop they arrive as plain text (v2.1.196 scheduled-fire rule + v2.1.215 removal of self-invocation).
- [claude-loops-scheduling] Harness-article re-read overturns prior model-version framing: the fix for premature completion is attributed to architecture, not to Opus 4.5 or 4.6.
- [claude-loops-scheduling] Methodology note: WebFetch's small-model summaries of the changelog were internally inconsistent (claimed oldest version was 2.1.200 while quoting 2.1.178); all version attributions in the claims were re-derived by downloading the raw CHANGELOG.md and grepping headings/line ...
- [claude-hooks-permissions] Changelog is now DATED (was undated at the 2026-07-20 pass): v2.1.210=Jul 14, v2.1.211=Jul 15, v2.1.212=Jul 17, v2.1.214=Jul 18, v2.1.215=Jul 19, v2.1.216=Jul 20, v2.1.217=Jul 21, v2.1.218=Jul 22, latest v2.1.220=Jul 25 2026. This resolves the prior open question about und ...
- [claude-hooks-permissions] Auto-mode classifier default model moved to Claude Sonnet 5 for external sessions at v2.1.210 (Jul 14); the engineering post still says Sonnet 4.6 and was not updated, so its published rates (0.4%/17%/5.7% full pipeline) characterize Sonnet 4.6, not the current default.

- [claude-hooks-permissions] v2.1.211 (Jul 15) loosened the auto-mode push default to any branch of the working repo incl the default branch, and removed the 'Default / protected branches' environment context slot that treated main/master as protected; deploy-named branches (production/gh-pages) are s ...
- [claude-hooks-permissions] Hook-event count is resolved at 30 (SessionStart first, SessionEnd last) via a direct raw read of the guide lifecycle table plus two independent enumerations of the reference; the earlier 33-vs-31 conflict was an extraction artifact.
- [claude-hooks-permissions] ConfigChange blocking is confirmed to EXEMPT policy_settings (verbatim from the reference exit-code-2-per-event table), resolving the prior single-extraction uncertainty.
- [claude-hooks-permissions] In-window (2026-07-20+) auto-mode/hook changes at v2.1.218: useAutoModeDuringPlan defaults on (classifier reviews shell commands and protected-path writes during planning); rm -rf / and rm -rf ~ (incl in command/process substitution) moved from a fixed prompt to classifier ...
- [claude-hooks-permissions] v2.1.211 also fixed auto mode overriding a PreToolUse hook's `ask` decision for unsandboxed Bash — a hook `ask` now floors the decision at a prompt.
- [claude-hooks-permissions] v2.1.214 (Jul 18) fixed hooks with exit code 2 not blocking as documented when the hook's stdout JSON fails schema validation — relevant to the exit-code-2 blocking mechanic.
- [claude-hooks-permissions] No reversal found through v2.1.220 of either the Sonnet 5 classifier default or the push-to-any-branch default.
- [claude-context-memory-skills] Opus 5 landed: requires Claude Code v2.1.219+; on the Anthropic API the 'opus' and 'default' aliases now resolve to Opus 5 (was Opus 4.8 before v2.1.219). New full model id claude-opus-5.
- [claude-context-memory-skills] Subagent limit machinery expanded since the prior horizon: NEW concurrent limit of 20 (CLAUDE_CODE_MAX_CONCURRENT_SUBAGENTS, v2.1.217+) and NEW configurable nesting depth (CLAUDE_CODE_MAX_SUBAGENT_SPAWN_DEPTH, v2.1.217+) whose default moved from 5 (unchangeable, <=v2.1 ...
- [claude-context-memory-skills] Auto memory gained a 'modified' ISO-8601 frontmatter timestamp field, written on the next write to any file that already has frontmatter, requiring v2.1.214+.
- [claude-context-memory-skills] Skills changes: /verify and /code-review are now user-invocation-only (v2.1.215) and can't be preloaded into subagents; boolean frontmatter now accepts yes/no/on/off/1/0 (v2.1.218); /code-review runs as a forked subagent (v2.1.218); forked skills can set background:fal ...
- [claude-context-memory-skills] Subagent output scanning (backslash insertion + instruction-shaped marker line on subagent reports) requires v2.1.210+ — a prompt-injection mitigation not previously captured.
- [claude-context-memory-skills] Confirmed the docs attribute the background-subagent permission-prompt-in-main-session behavior to v2.1.186 (resolving a prior conflict with a changelog mirror that said v2.1.208).

- [claude-context-memory-skills] Tooling limitation observed this pass: the env-vars page is large enough that the WebFetch markdown converter truncates before the M-Z portion, so negative ('ABSENT') findings from that fetch are unreliable — several definitely-present variables (e.g. CLAUDE_CODE_DISAB ...
- [codex-config-hooks] Stable moved 0.144.6 -> 0.146.0: 0.145.0 published 2026-07-21, 0.146.0 published 2026-07-29. Alpha line advanced to 0.147.0-alpha.2 (2026-07-30, tag rust-v0.147.0-alpha.2, prerelease=true).
- [codex-config-hooks] 0.145.0 (2026-07-21) shipped exactly one hook-mechanic change: a Windows fix for 'correctly quoted hook commands', alongside native exec-server sandboxing and network-proxy enforcement.
- [codex-config-hooks] 0.146.0 (2026-07-29) shipped NO hook/config.toml/AGENTS.md/trust/requirements.toml changes (focus: Agent Plugins manifests/marketplaces, thread forking, executor-provided skills, proxy/MCP fixes). So the core question -- did 0.144.6->stable change any hook event, handler type, t ...
- [codex-config-hooks] Documentation drift is the big change since 2026-07-20: the Hooks page now includes a per-event matcher table, per-event JSON output schemas (permissionDecision deny/allow, updatedInput rewriting, PermissionRequest decision.behavior/message), explicit hash-based trust language, ...
- [codex-config-hooks] config.toml reference now documents inline [hooks] tables, features.hooks (labeled stable/default-enabled), additionalContextLimit (default 2500; 0=full), commandWindows (Windows-only override), and projects.<path>.trust_level (trusted/untrusted gates project-scoped .codex/ laye ...
- [codex-config-hooks] Open question #1 (project_doc_max_bytes semantics) STILL unresolved and actively conflicting: AGENTS.md page = combined/accumulated cap (32 KiB default); config-advanced page = per-file ('how much to read from each AGENTS.md file').
- [codex-config-hooks] Open question #3 flipped in code: current main registry.rs CoreToolRuntime.pre_tool_use_payload default now returns Some for ALL Function-payload tool calls (None only for non-Function), which contradicts still-open issue #20204's premise that the default returns None and only s ...
- [codex-config-hooks] Open question #8: experimental_use_rmcp_client is confirmed ABSENT from both the official config-reference and config-advanced pages (tier-1 negative). Tier-4/5 aggregators claim it exists as [features] experimental_use_rmcp_client=true to enable the new RMCP client stack, but n ...
- [codex-exec-sandbox] Version cursor moved during the window: latest stable Codex CLI is now rust-v0.146.0 (2026-07-29), superseding rust-v0.144.6; rust-v0.147.0 is in alpha (rust-v0.147.0-alpha.2 dated 2026-07-30). The FOCUS version (0.144.6) is already one+ minor behind — re-baseline version-sensit ...
- [codex-exec-sandbox] GPT-5.6 context-window correction CONFIRMED at tier 1: rust-v0.144.6 (2026-07-18) corrected GPT-5.6 Sol/Terra/Luna bundled context windows to 272,000 tokens (PRs #33972, #34009). Tier-3/4 aggregators frame this as a reduction from a previously higher advertised window (~372K); t ...

- [codex-exec-sandbox] Linux sandbox backend change: the codex-rs linux-sandbox README now presents Bubblewrap (bwrap) + seccomp as the DEFAULT enforcement, with Landlock+mount protection demoted to a legacy fallback (features.use_legacy_landlock=true). Prior passes appear to have assumed Landlock+sec ...
- [codex-exec-sandbox] rust-v0.146.0 in-scope changes newer than 2026-07-20: (a) 'Detach non-interactive subprocesses from stdin' (#34612) — affects codex exec/headless behavior; (b) 'Retain more available skills under tight context budgets and warn when skill catalogs must be truncated' (#34732/#3473 ...
- [codex-exec-sandbox] Approval value set clarified: --ask-for-approval = untrusted | on-request | never, and approval_policy enum = untrusted | on-request | never | {granular}. 'on-failure' is absent from both the current CLI reference and the config-reference enum. If any prior repo note listed on-f ...
- [codex-exec-sandbox] Year-transcription hazard: the GitHub release tag pages intermittently render/transcribe the copyright year as 2024, but the releases index dates the same tags in 2026 (0.136.0=2026-06-01, 0.144.6=2026-07-18, 0.146.0=2026-07-29). Prior open question on rust-v0.144.6's year resol ...
- [security-sandboxing-injection] [security-sandboxing-injection] RESOLVES prior OQ 58. The Claude Code sandbox settings surface advanced past the v2.1.212 marker ceiling noted on 2026-07-20. As of 2026-07-30 the sandboxing docs expose min-version markers through v2.1.219 and the latest release is v2. ...
- [security-sandboxing-injection] [security-sandboxing-injection] NETWORK-ISOLATION re-verification: claims 01 and 04 re-checked verbatim against the live tier-1 sandboxing docs on 2026-07-30 and STAND unchanged - Seatbelt (macOS), bubblewrap+socat (Linux/WSL2), optional seccomp via `@anthropic-ai/san ...
- [security-sandboxing-injection] [security-sandboxing-injection] RESIDUAL-ATTACK-RATE re-verification: claims 13/14/15 re-checked against live arXiv abstracts on 2026-07-30 and STAND unchanged - Nasr et al. 2510.09023 (12 defenses, adaptive ASR >90% for most vs near-zero originally), PromptArmor 2507 ...
- [security-sandboxing-injection] [security-sandboxing-injection] RESOLVES prior OQ 64 (local platform). `wsl -l -v` on this machine lists three WSL2 distributions - Ubuntu, docker-desktop, and AgenticBench - all VERSION 2. So the Claude Code built-in bubblewrap sandbox CAN run on this estate under WS ...
- [security-sandboxing-injection] [security-sandboxing-injection] OWASP flagship unchanged: no new OWASP GenAI primary published after 2026-07-20. The genai.owasp.org resources page lists State of Agentic AI Security and Governance v2.01 (2026-06-01) as newest; OWASP Top 10 for Agentic Applications fo ...
- [security-sandboxing-injection] [security-sandboxing-injection] Sophos assigned source is dated 2026-05-12 (Ross McKerchar, Sophos CISO) - PRE-window, so not new material this cycle; it is being reviewed for the first time out of pending-review. Companion Sophos posts ('The OpenClaw experiment is a ...

- [security-sandboxing-injection] [cross-bucket / model-refresh] The Claude Code changelog also lists, under v2.1.219 (2026-07-24), a new default Opus model id and a change raising default subagent nesting depth (with an env var to disable), and under v2.1.216+ several non-sandbox items. These are out ...
- [evals-judging] METR: the assigned 2025-07-10 study is now superseded by METR's 2026-02-24 update (metr.org/blog/2026-02-24-uplift-update/), which reports the follow-up experiment was undermined by selection bias (developers refusing the no-AI arm; pay cut $150->$50/hr) and gives confounded new esti ...
- [evals-judging] SWE-bench landscape: the family now includes a 'Multilingual' variant, and Scale AI's SWE-bench Pro (1,865 tasks; public/held-out/commercial splits; GPL + private codebases for contamination resistance) has emerged as the harder successor. Multiple tier-4/5 aggregators state OpenAI d ...
- [evals-judging] CodeScaleBench: frozen published suite csb-v1-mixed371 (371 tasks) diverges from a churning working tree (repo ops notes cite ~550 task.toml files and ~274 canonical/active tasks with verification_mode/use_case_category schema gaps). Pin a commit when reproducing; treat 371 (151 SDLC ...
- [evals-judging] OpenHands Index: launched 2026-01-29 with 9 models across 5 execution-based benchmarks; tier-4/5 aggregators report growth to ~28 model variants and a May-2026 snapshot led by Claude Opus 4.7 (Adaptive), but those per-model scores are not confirmed on any primary source.
- [evals-judging] Model-identifier verification (partial resolution of prior open question #7): confirmed on PRIMARY sources this pass are GPT-5.4, Gemini-3.1 Pro Preview, Claude Opus 4.7, Claude Sonnet 4.6, Kimi K2.6, GLM-5.1, DeepSeek V4 Pro/Flash, Seed 2.0 Pro, Qwen3.6 Max Preview, MiMo v2.5 Pro (R ...
- [academic-deep-dive] Google blog vs arXiv discrepancy: the assigned Google Research blog reports 180 configurations, 17.2x/4.4x error amplification, and 4 benchmarks (Finance-Agent, BrowseComp-Plus, PlanCraft, Workbench), but arXiv 2512.08296 'Towards a Science of Scaling Agent Systems' (v1 2025-12- ...
- [academic-deep-dive] Assigned coding-validity paper 2607.01211 advanced to v2 on 2026-07-16 (v1 2026-07-01); all extracted numbers reflect v2.
- [academic-deep-dive] Related coding-benchmark-validity position paper 2606.17799 advanced to v2 on 2026-07-18 (v1 2026-06-16).
- [academic-deep-dive] No assigned source received a version bump AFTER the prior run date (2026-07-20): the newest assigned items are 2607.05904 (v1 2026-07-07) and 2607.01211 (v2 2026-07-16), both predating it. This bucket's novelty is the deep numeric extraction plus newly surfaced citing/adjacent ...
- [academic-deep-dive] New cross-source tension surfaced this pass: the SIGN of multi-agent error propagation appears topology-dependent - arXiv 2606.07937 measures net attenuation (amplification factor 0.644) in 3-agent sequential-refinement chains, directly opposite the Google blog's 17.2x amplifica ...

- [academic-deep-dive] New coding-specific reward-hacking evidence cluster discovered adjacent to the assigned six: 2606.26300 (Verification Horizon, hacked-resolved 28.57%->0.56% via judge+trajectory monitoring), 2605.21384 (SpecBench, ~28pp visible/holdout gap growth per 10x code size), and 2607.087 ...
- [youtube-lane] Claude Opus 5 released 2026-07-24 as the new Opus-tier flagship (succeeds Opus 4.8) - post-prior-run (Jul 20) drift; tier-1 anthropic.com/news/claude-opus-5.
- [youtube-lane] Codex CLI 0.145.0 (Jul 21) and 0.146.0 (Jul 29) shipped post-Jul-20: Multi-agent V2 with configurable sub-agent models/reasoning levels, `/import` of Claude Code + Cursor settings/MCP/plugins/sessions, Amazon Bedrock login with GPT-5.6 Sol as default Bedrock model, Agent Plugins manif ...
- [youtube-lane] OpenAI ChatGPT desktop 26.715 (Jul 23) added GPT-Live voice across Chat/Work/Codex and macOS 'appshots' screen context; 'Sign in with ChatGPT' beta rolled out Jul 29 (partners: Airtable, GitLab, HubSpot, Notion, Supabase, Vercel).
- [youtube-lane] New in-window video: Latent Space published 'Codex from 0 to 10M Users: Building ChatGPT Work' (Akshay Nathan, OpenAI) on 2026-07-28.
- [youtube-lane] Claude Code tier-1 whats-new for July weeks 27-29 (v2.1.195-v2.1.212) documents: Claude Sonnet 5 as new default (native 1M context, adaptive thinking on by default), Claude in Chrome GA, subagents run in background by default, in-app Desktop browser, `/doctor` (+ `/checkup` alias), `/f ...
- [youtube-lane] Context note: GPT-5.6 family GA (Sol/Terra/Luna) was Jul 9 and AIEWF talk uploads clustered ~Jul 15 - both in-window (since ~Jul 1) but predate the Jul 20 prior run, so they are valid candidates rather than post-run drift.
- [reddit-lane] OPERATIONAL: reddit.com is blocked to the WebSearch/WebFetch user agent (confirmed via API 400 'domains not accessible' on allowed_domains=[reddit.com], and site:reddit.com queries return only non-Reddit results). This discovery lane cannot fetch Reddit threads directly; it must corrob ...
- [reddit-lane] NEW since 2026-07-20: Claude Opus 5 (claude-opus-5) shipped in v2.1.219 on 2026-07-24 as the new default Opus model (1M context, fast mode $10/$50 per MTok); Opus 4.7 removed from fast mode.
- [reddit-lane] Subagent nesting default reversed twice in-window: disabled by default in v2.1.217 (2026-07-21), then re-enabled to depth 3 by default in v2.1.219 (2026-07-24) — effective behavior is patch-version-dependent.
- [reddit-lane] Codex CLI advanced to 0.146.0 stable (2026-07-29) and introduced an agent-plugin marketplace, creating a Codex-side third-party-plugin supply-chain surface paralleling Claude Code's plugin/marketplace risk.
- [reddit-lane] A concentrated batch of Claude Code Bash/permission-analyzer hardening landed 2026-07-11 to 2026-07-25 (v2.1.207-v2.1.220): glob allow-rule over-match (2.1.214), help/man and docker auto-approve gaps (2.1.214), compound-&&/redirect and Windows-network-path checks (2.1.216), hook exit-c ...

- [reddit-lane] Both CVEs cited (CVE-2026-25723 fixed 2.0.55; CVE-2026-33068 fixed 2.1.53) are patched and predate the ~3-week window; they are included as durable, tier-1/2-verifiable footgun THEMES (command-chaining sandbox escape; repo-controlled settings flipping permission mode), not as current-v ...
- [reddit-lane] The 'Friendly Fire' brief (AI Now, 2026-07-08) tested Claude Code only up to 2.1.199 and Codex 0.142.4; current versions (Opus 5 / v2.1.219+, Codex 0.146.0) were not in its matrix, so applicability to current builds is unverified.
- [model-catalog-refresh] MAJOR (Anthropic): Claude Opus 5 (`claude-opus-5`) launched and is now the recommended flagship; the `opus` alias resolves to Opus 5 on the Anthropic API (was Opus 4.8 in the 2026-07-19 catalog). Opus 4.8 is now a legacy model. Requires Claude Code v2.1.219+ (was v2.1.154 for ...
- [model-catalog-refresh] Anthropic: the account-type `default` and Claude Code default now resolve to Opus 5 (Max/API tiers) and Sonnet 5 (Pro tiers); prior catalog implied Opus 4.8 as the top Anthropic model.
- [model-catalog-refresh] Anthropic: new `best` alias (Fable 5 where available, else latest Opus) and new content-based automatic fallback (Fable 5 biology->Opus 5 / cybersecurity->Opus 4.8; Opus 5 cybersecurity->Opus 4.8, biology->refusal), both v2.1.219+.
- [model-catalog-refresh] Anthropic effort: default is `high` on all effort models except Opus 4.7 which defaults to `xhigh` (prior catalog stated `high` universally). Opus 4.6 and Sonnet 4.6 support `max` but not `xhigh`.
- [model-catalog-refresh] OpenAI (CENTRAL DISCREPANCY): the assigned FOCUS flagged a 'GPT-5.6 272k correction', but the current official API models page (raw text, confirmed via browser) shows Context window '1.05M' for all three GPT-5.6 models; no official page (models, changelog, config-reference) s ...
- [model-catalog-refresh] OpenAI: GPT-5.6 Sol/Terra/Luna are the recommended Codex family (Sol flagship at $5/$30, Terra $2.50/$15, Luna $1/$6; all 128K max output, Feb 16 2026 cutoff). The prior catalog's capability-selector resolution notes (luna/terra/sol) remain correct.
- [model-catalog-refresh] OpenAI: gpt-5.2 and gpt-5.3-codex are now deprecated in Codex (ChatGPT sign-in); gpt-5.5 remains a selectable older generation and is still the illustrative `model` example in the config reference; gpt-5.1 is no longer referenced. Resolves prior open question #7.
- [model-catalog-refresh] OpenAI docs URL moves: platform.openai.com/docs/models -> 301 -> developers.openai.com/api/docs/models; developers.openai.com/codex/changelog -> 308 -> learn.chatgpt.com/docs/changelog. Resolves prior open question #4 for the Codex changelog canonical URL.
- [model-catalog-refresh] OpenAI effort vocabulary now differs across three surfaces: API models page (none/low/medium/high/xhigh/max), config.toml model_reasoning_effort (minimal/low/medium/high/xhigh), and ChatGPT/Codex app slider (Light or Low, Medium, High, Extra High, Max, Ultra). The catalog's e ...
- [model-catalog-refresh] Catalog freshness: Models/providers.json generated 2026-07-19 and expired 2026-07-26; a candidate patch should refresh generated_at/expires_at and apply claims 01-33.

## Proposed repository changes

Full diff plan in `proposed-changes.md`: (0) security-review gate on the flagged bucket; (1) candidate model-catalog refresh; (2) pin Claude Code v2.1.220 / Codex 0.146.0 and correct last week's version attributions; (3) reflect the Reddit-surfaced permission footguns; (4) apply the source patch; (5) fold the academic references into the knowledge base and diff CodeScaleBench's admission triad against `tools/adaptive`.

## Eval and ablation plan

`eval-plan.json`: 8 cases (7 blocking), **zero provider spend**. Blocking gates include **EV-SEC** (security-flagged bucket human review) and **EV-MODEL** (re-verify every model fact before any catalog write).

| Case | Name | Blocking |
|---|---|---|
| EV-SEC | Security-flagged bucket human review | yes |
| EV-01 | Source patch applies and is reversible | yes |
| EV-02 | No stable doc cites the retired URL | yes |
| EV-MODEL | Model-catalog re-verification before catalog refresh | yes |
| EV-COR | Corrected claims reconciled with contracts | yes |
| EV-VS | Version-sensitive sample re-verification | yes |
| EV-REJ | Rejected claims stay out of guidance | yes |
| EV-RPT | Report source integrity + PDF URIs | no |

## Expected impact

- The model catalog gets a verified refresh path (Opus 5 flagship, Opus 4.8 legacy, GPT-5.6 line, effort-level correction incl. the `ultracode` clarification).
- Last week's version ambiguities are resolved against the raw changelog (v2.1.220; the v2.1.212 caps; the /loop-resume fix at v2.1.211).
- The permission guidance gains two concrete, corroborated footguns to avoid.
- The knowledge base gains 9 external papers that quantitatively corroborate the repo's diff-derived-oracle direction.
- No behavioural change until promotion; contracts and Stable untouched.

## Risks

- **One bucket is under security review** (24 claims) - do not promote them without EV-SEC sign-off.
- **Model facts are the most volatile** in the set; re-verify at promotion (EV-MODEL), never fabricate.

- **244 version-sensitive claims** carry expiry dates; a late promotion must re-check.
- **43 newly discovered sources** entered on a single pass; re-check at first due date.
- Reddit/YouTube claims are leads; several were corroborated to the changelog, but confidence is only as good as that corroboration.

## Rollback

Everything is confined to `Research/candidate-packages/2026-07-30-weekly-research/`; delete that directory to abandon. No other repo state changed; no commit or push. If promoted, rollback restores pre-patch `registry.json` and `providers.json` from the promotion rollback manifest (EV-01 requires a hash-exact restore first).

## Open questions

74 carried forward.

- [claude-loops-scheduling] ScheduleWakeup's full input schema is still only partially documented: the tool-table row now pins stop:true (v2.1.202+), the 1min-1hr firing window, and session_crons surfacing in Stop hook input, but the delay-input field NAME and UNITS, and whether an explicit absolute n ...
- [claude-loops-scheduling] The /goal evaluator's per-turn token cost is still unquantified - docs only say it is 'typically negligible compared to main-turn spend.' No published number; would need local measurement to attach a figure.
- [claude-loops-scheduling] Routines research-preview cap NUMBERS remain unpublished: the per-routine and per-account hourly GitHub-webhook caps and the daily per-account routine-run cap are visible only in the authenticated UI at claude.ai/code/routines. Capacity planning still requires reading the U ...
- [claude-loops-scheduling] No tier-1/tier-2 source exists for the circulating token-multiplier figures (single-agent ~4x standard chat, multi-agent ~15x); the Anthropic harness article provides no multipliers. A loop-overhead cost number likely has to be measured locally rather than cited.
- [claude-loops-scheduling] Blog 'Loop engineering: Getting started with loops' publication date is only provisionally confirmed: a direct page fetch reads June 30, 2026, but web search could not independently corroborate it and the earlier July 7, 2026 retrieval was not reproduced. Re-verify in-page ...
- [claude-hooks-permissions] Firing order (residual): the docs state PreToolUse fires before any permission-mode check and the lifecycle diagram depicts PreToolUse -> PermissionRequest, but there is still no standalone normative sentence explicitly asserting the intra-tool-call ordering between the Pr ...
- [claude-hooks-permissions] Whether hooks defined in skill or agent frontmatter share the exact deny-beats-bypassPermissions precedence as settings-file hooks: the guide's precedence statement addresses 'PreToolUse hooks' generically and is not explicitly scoped by configuration source. v2.1.218 adds ...

- [claude-hooks-permissions] v2.1.213 was not surfaced in this changelog fetch (gap between v2.1.212 dated Jul 17 and v2.1.214 dated Jul 18); its exact date and contents are unconfirmed this pass.
- [claude-hooks-permissions] The Sonnet 5 classifier's real-traffic false-positive/false-negative rates are unpublished; only the older Sonnet 4.6 numbers exist, so whether classifier accuracy changed with the v2.1.210 model default is unmeasured.
- [claude-context-memory-skills] Residual on the 200K auto-compaction threshold: CLAUDE_AUTOCOMPACT_PCT_OVERRIDE confirms a 1-100 percentage knob and the docs say standard 200K models compact 'at the 200K boundary', but NO numeric DEFAULT percentage is published for the standard case. The practitioner ...
- [claude-context-memory-skills] 'Microcompact' still could not be confirmed as an official tier-1 product name or env var. Tier-3/4 community sources describe the behavior as 'microcompaction' (hot-tail inline results, older tool outputs offloaded to disk by reference), but no tier-1 DISABLE_MICROCOM ...
- [claude-context-memory-skills] MAX_MCP_OUTPUT_TOKENS: presence/absence on the current env-vars page remains UNVERIFIED because the WebFetch converter truncated before the M-Z range; do not assert it is gone. Re-check via section anchor or raw markdown before relying on or retiring any repo reference ...
- [claude-context-memory-skills] No first-party token multiplier exists for agent teams running in NORMAL (execution) mode; the ~7x figure is explicitly plan-mode-only, leaving the cost ladder unquantified for the mode where most real work happens.
- [claude-context-memory-skills] Still no coding-specific measurement of Claude Code agent teams versus a single session anywhere in first-party docs; the token-cost guidance is qualitative ('significantly more', 'scales linearly with team size').
- [claude-context-memory-skills] Unresolved changelog item carried forward (not in this bucket's assigned sources): the reported fix for the context window / auto-compact indicator briefly resetting to 200K after a CLI auto-update was not located this pass.
- [claude-context-memory-skills] Not chased this pass (outside assigned sources): direct verification of the suspicious shared 2026-06-03 publication dates on the Anthropic Skills post and the tembo.io post.
- [codex-config-hooks] project_doc_max_bytes semantics remain contradictory across two official pages: AGENTS.md page (combined/accumulated cap, 32 KiB default) vs config-advanced page (per-file, 'each AGENTS.md file'). Needs source-level or empirical resolution of which is the true runtime behavior.
- [codex-config-hooks] Exact Codex version that stabilized hooks / made features.hooks default-enabled / introduced inline [hooks] tables and requirements.toml managed hooks is still not pinned. CHANGELOG.md in-repo just redirects to the releases page, and 0.145.0/0.146.0 notes do not announce it, so ...
- [codex-config-hooks] Does 0.146.0 STABLE contain the broadened PreToolUse coverage seen on main (registry.rs default returns Some for all Function payloads), or does it still ship the limited

shell/unified_exec/apply_patch/mcp set from issue #20204? Requires reading source at the rust-v0.146.0 tag o ...

- [codex-config-hooks] Per-event UNSUPPORTED hook output fields are still not enumerated field-by-field (e.g. whether continue, stopReason, and suppressOutput are honored for PreToolUse vs ignored). Docs list common fields but not the per-event exclusions.
- [codex-config-hooks] The hook trust hash is described as covering the 'exact hook definition' / 'current hash', but the docs do not itemize whether the hash includes the matcher and timeout (vs only the command string), so whether editing a matcher silently untrusts a hook is implied but not explici ...
- [codex-config-hooks] experimental_use_rmcp_client: not present in the official config-reference or config-advanced pages as of 2026-07-30. Tier-4/5 sources assert it is set under [features] to enable the new RMCP/remote-MCP client stack; needs a tier-1 primary (official docs or repo source) to confi ...
- [codex-config-hooks] 0.147.0-alpha.1 and 0.147.0-alpha.2 release-note bodies failed to render via fetch (GitHub 'error while loading'); the alpha line's hook/config/trust delta beyond 0.146.0 is unconfirmed for this pass.
- [codex-exec-sandbox] Default value of approval_policy: the config reference enumerates the enum (untrusted|on-request|never|granular) but does not state the default; the approvals page hints 'on-request' is default but not crisply. Needs a tier-1 statement before any default is documented.
- [codex-exec-sandbox] Default value of MCP default_tools_approval_mode: enum (auto|prompt|writes|approve) is tier-1 confirmed but the default is unstated in both the MCP page and config reference. This governs the security posture of an unconfigured MCP server — do not assert until confirmed.
- [codex-exec-sandbox] Was 'on-failure' ever a valid --ask-for-approval / approval_policy value, and if so in which release was it removed? It is absent from current tier-1 references; the removal (vs never-existed) is unverified.
- [codex-exec-sandbox] Physical location/mechanism of archived sessions: 0.136.0 release notes confirm the resume/fork protection but not whether archiving MOVES files to an archived_sessions/ directory or only sets a metadata flag. Only tier-4/5 (danielvaughan KB) asserts ~/.codex/archived_sessions/; ...
- [codex-exec-sandbox] Are Codex session/rollout files created world-readable (mode 0644) on Unix? Asserted at tier 5 only; no tier-1/2 evidence located this pass. If true it is a real operational-security finding; if false it is a fabrication to flag. Requires direct verification of the rollout-write ...
- [codex-exec-sandbox] Do skills load and function inside `codex exec` (non-interactive), and does implicit invocation apply there? The build-skills page still does not address non-interactive mode. Materially affects whether skills are a viable automation-packaging mechanism vs interactive-only.
- [codex-exec-sandbox] Does the exec JSONL stream define stable, enumerated exit codes? Docs only state Codex exits non-zero on failure; no exit-code table located in the non-interactive page or CLI reference. CI wrappers still lack a documented code contract.

- [codex-exec-sandbox] Does agents.<name>.config_file exist and can it point at agent definitions outside .codex/agents/ (a supply-chain surface, since agent files can override sandbox_mode)? Not found in the config reference or subagents page this pass.
- [codex-exec-sandbox] Has the auto-compaction regression (#16033) been fixed in any 0.14x release? Not verified; 0.145.x release notes were not deep-read and 0.146.0 notes did not mention it.
- [codex-exec-sandbox] features.use_legacy_landlock is documented in the linux-sandbox README but did not surface in the general config reference. Confirm it is a live, supported [features] key (vs README-only) and where it is canonically documented.
- [codex-exec-sandbox] Did rust-v0.145.x introduce any in-scope sandbox/exec/approval/MCP changes? Only rust-v0.146.0 release notes were deep-read this pass; 0.145.x should be reviewed next pass to complete the drift picture.
- [codex-exec-sandbox] CBSE ('Configuration-Based Sandbox Escape') and CVE-2025-59532 sandbox-escape claims remain UNCORROBORATED by any tier-1/2 source or OpenAI security advisory. Do not assert; investigate provenance before any Codex threat-model use.
- [codex-exec-sandbox] Review-queue behavior for runs with NO findings: does the automations UI auto-archive empty runs? The automations page documents unread indicators + manual mark-all-as-read but did not confirm auto-archiving of empty runs (tier-3 claim only).
- [security-sandboxing-injection] [security-sandboxing-injection] (carry forward, OQ 62) The Google April-2026 Common Crawl '32% increase in malicious attempts between November 2025 and February 2026' (cited by Sophos, t3) and the '340% year-over-year' prompt-injection surge (cited by tier-5 aggregato ...
- [security-sandboxing-injection] [security-sandboxing-injection] (carry forward, OQ 61) Third-party tools named in the Sophos post (nono, Agent Vault, cedar-for-agents, Kelos, LlamaFirewall) remain unverified for existence and maintenance; the GitHub org/repo paths returned by the summarizer are not ...
- [security-sandboxing-injection] [security-sandboxing-injection] (carry forward, OQ 60) amux.io was NOT re-tested this pass, so the cause of its earlier 403 (bot/UA filtering vs geo-block vs removal) is still unknown; the retire decision for that tier-5 source is unconfirmed.
- [security-sandboxing-injection] [security-sandboxing-injection] (carry forward, OQ 63) Northflank's '97ms median time-to-interactive / 100% success' ComputeSDK benchmark was not re-examined this pass (off the sharp prompt-injection/trifecta focus); whether ComputeSDK is vendor-affiliated and whether ...
- [security-sandboxing-injection] [security-sandboxing-injection] NEW lead for next cycle: NetInjectBench (arXiv 2607.10490, July 2026) surfaced as an in-window indirect-prompt-injection benchmark for tool-using LLM agents in network operations, but was not primary-verified this pass and is domain-nar ...
- [security-sandboxing-injection] [security-sandboxing-injection] The '88% of organizations reported confirmed or suspected AI agent security incidents' figure appeared only in a tier-5 search summary with no located primary survey; do not assert it.

- [security-sandboxing-injection] [security-sandboxing-injection] Claim 12 (NVIDIA three sandboxing controls, t3, 2026-01-30) was NOT re-fetched this pass; it is not version-sensitive and is presumed stable, but a direct re-verification is due at its next tier-3 recheck to confirm the egress/deny-list ...
- [evals-judging] Prior #5 unresolved: whether the self-play de-anchoring mitigation (commit-before-judging, GSM8K false positives 0.719->0.012) transfers to open-ended coding review, where the judge cannot cheaply produce its own reference solution, was not addressed this pass; carry forward.
- [evals-judging] Prior #6 unresolved (academic bucket): the Imaging-101 (57 tasks, Chen/Ying/Sun) vs ImagingBench/arXiv:2607.07189 (20 tasks, Chung et al.) naming conflict was not investigated in this eval-judging pass.
- [evals-judging] Prior #7 partially resolved: 'Claude Mythos 5', 'Claude Fable 5', and 'GPT-5.6 Sol' still appear only in tier-4/5 aggregators and are unconfirmed on any primary source; swebench.com's aggregator-reported top score (~95.5% for 'Claude Mythos 5') could not be verified because the live ...
- [evals-judging] Prior #9 unresolved: JudgeBiasBench (frontier judges reportedly >50% error on advanced bias tests) — the primary source was again not located; targeted lookup still needed.
- [evals-judging] NEW: OpenAI's primary write-up 'Why we no longer evaluate SWE-bench Verified' (openai.com/index/why-we-no-longer-evaluate-swe-bench-verified/) returned HTTP 403 via WebFetch twice; its specific figures (a ~27.6% audited subset, >=59.4% of audited problems having flawed tests that rej ...
- [evals-judging] NEW: On Scale's SWE-bench Pro public leaderboard, entries labeled 'Muse Spark 1.1' (61.50%) and 'Muse Spark' (55.00%) rank at/above gpt-5.4 (xHigh); whether 'Muse Spark' is a real model, a vendor submission alias, or a Scale-internal label is unresolved and the identifier could not b ...
- [evals-judging] Prior #10 narrowed but open: RuVerBench (843 human-labeled agentic-coding trajectory rubrics) and CodeJudgeBench (test-based code correctness over generation/repair/test-gen) now cover adjacent ground, but a benchmark that judges code-DIFF correctness against HUMAN gold labels specif ...
- [evals-judging] NEW: RuVerBench's majority-voting saturation is documented qualitatively ('within three to five votes'); the exact per-k BAcc deltas from its Figure 6 were not extracted numerically this pass and could sharpen the recommended k if needed.
- [evals-judging] NEW (discovery lead, not asserted): Berkeley RDI reported (April 2026, rdi.berkeley.edu/blog/trustworthy-benchmarks/) an automated agent that found exploitable scoring mechanisms in every major agent benchmark tested (SWE-bench, WebArena, OSWorld, GAIA); and reward-hacking benchmarks ...
- [academic-deep-dive] RuVerBench (2606.29920) top-model accuracy/F1 and the AgenticCoding-vs-DeepResearch split accuracy were not retrievable: arXiv HTML returned 404 for v1 and no-version, and neither the abstract nor the repo README states a headline accuracy. Needs a PDF pass next run to get the l ...
- [academic-deep-dive] arXiv 2606.01066 'Before the Model Learns the Bug: Fuzzing RLVR Verifiers' (2026-05-31, Jaideep Ray) advertises false-positive/false-negative/disagreement/exploit/uncertainty

metrics but no concrete rates were retrievable from the abstract page; a fuzzing-based verifier-admissio ...

- [academic-deep-dive] RHB (2605.02964) mid-table per-model exploit rates (o3 11.8%, o4-mini 8.4%, o3-mini 7.1%, o1 6.8%, Gemini 2.5 Pro 4.6%, Claude 3.7 Sonnet 3.9%) came from an HTML-table summarization; only the abstract-confirmed extremes (0% Sonnet 4.5, 13.9% DeepSeek-R1-Zero, 0.6% DeepSeek-V3) w ...
- [academic-deep-dive] Whether arXiv 2512.08296 is the preprint of the Google 'science of scaling agent systems' blog, a superseded version, or a different study, and which figure set is authoritative, is unresolved (see drift note).
- [academic-deep-dive] 2606.26300's Table 3 hacked/clean-resolved numbers are stated as 'average across benchmarks'; whether the average over SWE-Bench Verified/Multilingual/Pro is task-weighted or benchmark-weighted was not confirmed.
- [academic-deep-dive] Semantic Scholar / Google Scholar forward-citation enumeration of the assigned six could not be completed (Semantic Scholar search page returned only navigation chrome to WebFetch); citing-paper discovery this pass relied on WebSearch and is likely incomplete - a dedicated citat ...
- [youtube-lane] Did Anthropic's official @anthropic-ai YouTube channel post a dedicated Claude Opus 5 (Jul 24, 2026) launch video, or was in-window Opus 5 video coverage only third-party? Could not confirm an official video this pass.
- [youtube-lane] Lance Martin's AIEWF 2026 talk 'Claude for Long-Horizon Tasks': do the third-party-attributed mechanics ('brain/hands' decoupling, offline 'dreaming' memory correction) actually appear in the talk? The tier-1 Anthropic long-running-Claude research page (Mar 2026) documents a different ...
- [youtube-lane] How do Anthropic's two model-naming schemes relate - Fable/Mythos (Fable 5, Mythos 5) vs Opus/Sonnet/Haiku (Opus 5, Sonnet 5)? Sources conflict on hierarchy (e.g., 'cost-efficient version of Fable 5'); do not assert a mapping until a tier-1 model-overview page clarifies it.
- [youtube-lane] Content of the OpenAI Codex AIEWF 2026 workshop 'Full Workshop: Setting Yourself Up for Success' (Jason Liu) - not yet reviewed; likely a concrete source of Codex best-practice workflow guidance for a future pass.
- [youtube-lane] Codex/ChatGPT Work user-scale figures from the Jul 28 Latent Space episode (~10M combined within 2 weeks; >5M Codex WAU in June; knowledge workers ~20% growing 3x faster) are guest-stated with no tier-1 corroboration found - verify against any official OpenAI usage disclosure.
- [youtube-lane] Unverified t4 claim that 'GPT-5.6 Sol Ultra' shipped in the Codex client around Jul 6, 2026 - not found in the official Codex changelog; confirm or drop.
- [reddit-lane] Unverified community signal (widely repeated on HN/Reddit per secondary aggregators like firecrawl and developersdigest): 'Claude Code commits/iterates fast; Codex reasons longer and is more thorough on complex tasks.' This is subjective and not atomically verifiable from a tier-1/2 so ...

- [reddit-lane] Does current Codex CLI 0.146.0 still fail to auto-load a global ~/.codex/AGENTS.md? The only source (openai/codex issue #8759) is on 0.77.0 and closed 'not planned'; needs a fresh repro on 0.146.x before any claim.
- [reddit-lane] Magnitude of the cost surprise from Sonnet 5 'adaptive thinking on by default' + 1M-token context: no verified token/dollar figure was found — only anecdote. Needs a controlled /usage measurement to assert.
- [reddit-lane] Whether the Friendly Fire planted-binary vector still triggers on Claude Opus 5 (v2.1.219+) and Codex 0.146.0, given the intervening prompt-injection hardening (e.g. v2.1.210 'Hardened the Agent tool against indirect prompt injection via content a subagent read'). Untested.
- [reddit-lane] Direct Reddit corroboration for the auto-continue backlash beyond the cited GitHub issue #73125 (~384 upvotes) could not be obtained because reddit.com is inaccessible to the crawler; the community-sentiment magnitude on r/ClaudeAI is unverified.
- [reddit-lane] Is there an official Anthropic security advisory (GHSA) or CVE for the marketplace-plugin hijack class described by PromptArmor, or does Anthropic consider it out-of-scope (as OpenAI/Anthropic reportedly did for Friendly Fire)? No vendor advisory was located; only the vendor hooks docs ...
- [model-catalog-refresh] GPT-5.6 context window: reconcile the FOCUS's '272k correction' against the current official figure of 1.05M on developers.openai.com/api/docs/models. Determine whether 272k was a stale prior value now superseded by an expansion to 1.05M, or whether it refers to a non-API sur ...
- [model-catalog-refresh] Is there a tier-1 Anthropic announcement confirming the commercial/subscription-inclusion terms for Fable 5 (prior open question #6)? Model existence, GA date (2026-06-09), CLI gate (v2.1.170), ZDR restriction, and the `best` alias are tier-1 confirmed, but the specific subsc ...
- [model-catalog-refresh] Anthropic parameter constraint: verify the prior catalog claim that temperature/top_p/top_k return HTTP 400 on Opus 4.7+ and Sonnet 5 against the model-deprecations page (not re-fetched this run); confirm whether it also applies to Opus 5.
- [model-catalog-refresh] Carry-forward (out of model-catalog scope, still unresolved): documented default of CLAUDE_CODE_OTEL_CONTENT_MAX_LENGTH (5000 vs 60KB) (#1); env-vars-reference presence of CLAUDE_CODE_MAX_WEB_SEARCHES_PER_SESSION / MAX_SUBAGENTS_PER_SESSION / FORWARD_SUBAGENT_TEXT / FORK_SUBA ...

## Sources

123 unique sources, each once with tier and access date.

**Tier 1** (63)


- Agent approvals & security | ChatGPT Learn - accessed 2026-07-30
- Anthropic Engineering — Claude Code auto mode classifier deep dive - accessed 2026-07-30
- anthropics/claude-code CHANGELOG.md (raw, main) - accessed 2026-07-30
- Automate work with routines (Routines docs) - accessed 2026-07-30

- Build skills | ChatGPT Learn - accessed 2026-07-30
- Claude Code documentation — Automate actions with hooks (hooks guide) - accessed 2026-07-30
- Claude Code documentation — Changelog - accessed 2026-07-30
- Claude Code documentation — Choose a permission mode - accessed 2026-07-30
- Claude Code documentation — Configure auto mode - accessed 2026-07-30
- Claude Code documentation — Hooks reference - accessed 2026-07-30
- Claude Code — What's new (Week 27) - accessed 2026-07-30
- Claude effort parameter - accessed 2026-07-30
- Claude models overview - accessed 2026-07-30
- Claude v1 discovery: github.com/openai/codex/blob/main/codex-rs/linux-sandbox/README.md - accessed 2026-07-30
- Claude v1 discovery: learn.chatgpt.com/docs/automations - accessed 2026-07-30
- Codex Advanced Configuration - accessed 2026-07-30
- Codex AGENTS.md documentation - accessed 2026-07-30
- Codex Changelog - accessed 2026-07-30
- Codex config.toml Configuration Reference - accessed 2026-07-30
- Codex Hooks documentation - accessed 2026-07-30
- Codex rust-v0.145.0 release notes - accessed 2026-07-30
- Codex rust-v0.146.0 release notes - accessed 2026-07-30
- codex-rs/linux-sandbox/README.md (raw) - accessed 2026-07-30
- Codex/ChatGPT models - accessed 2026-07-30
- Configure the sandboxed Bash tool - Claude Code docs - accessed 2026-07-30
- Create custom subagents — Run subagents in foreground or background / Available tools - accessed 2026-07-30
- Developer commands (CLI reference) | ChatGPT Learn - accessed 2026-07-30
- Effective context engineering for AI agents - accessed 2026-07-30
- Effective harnesses for long-running agents (Anthropic engineering) - accessed 2026-07-30
- Environment variables — CLAUDE_CODE_AUTO_COMPACT_WINDOW - accessed 2026-07-30
- Explore the context window — When your context fills up - accessed 2026-07-30
- Extend Claude with skills — Frontmatter reference / Skill descriptions are cut short - accessed 2026-07-30
- How Claude remembers your project — How it works - accessed 2026-07-30
- How the agent loop works (Agent SDK docs) - accessed 2026-07-30
- https://platform.openai.com/docs/models - accessed 2026-07-30
- Introducing Claude Fable 5 and Claude Mythos 5 — Claude Platform Docs - accessed 2026-07-30
- Keep Claude working toward a goal (/goal docs) - accessed 2026-07-30
- Loop engineering: Getting started with loops (Claude blog) - accessed 2026-07-30

- Manage costs effectively — When a developer asks about a limit - accessed 2026-07-30
- Model configuration — Sonnet 5 context window - accessed 2026-07-30
- Model Context Protocol | ChatGPT Learn - accessed 2026-07-30
- NIST SP 800-218A (Final) publication page - accessed 2026-07-30
- Non-interactive mode | ChatGPT Learn - accessed 2026-07-30
- npm @openai/codex latest manifest - accessed 2026-07-30
- NVD — CVE-2026-25723 - accessed 2026-07-30
- NVD — CVE-2026-33068 - accessed 2026-07-30
- OpenAI API models - accessed 2026-07-30
- OpenAI Codex changelog - accessed 2026-07-30
- openai/codex core/src/tools/registry.rs (main) - accessed 2026-07-30
- openai/codex GitHub Releases - accessed 2026-07-30
- openai/codex issue #20204 - accessed 2026-07-30
- openai/codex releases API - accessed 2026-07-30
- openai/codex-action (README) - accessed 2026-07-30
- openai/codex-action action.yml (raw) - accessed 2026-07-30
- Orchestrate teams of Claude Code sessions — Enable agent teams / Limitations - accessed 2026-07-30
- Release 0.136.0 · openai/codex - accessed 2026-07-30
- Release 0.144.6 · openai/codex - accessed 2026-07-30
- Run prompts on a schedule (/loop & scheduled-tasks docs) - accessed 2026-07-30
- Sandbox | ChatGPT Learn - accessed 2026-07-30
- sourcegraph/CodeScaleBench README - accessed 2026-07-30
- Subagents | ChatGPT Learn - accessed 2026-07-30
- THU-KEG/RuVerBench README - accessed 2026-07-30
- Tools reference (WebSearch session search limit) - accessed 2026-07-30

**Tier 2** (43)

- AI Now Institute — Friendly Fire: Hijacking Defensive Cyber AI Agents for RCE - accessed 2026-07-30
- Are Performance-Optimization Benchmarks (full text v2) - accessed 2026-07-30
- Are Performance-Optimization Benchmarks Reliably Measuring Coding Agents? (abstract) - accessed 2026-07-30
- Auditing Reward Hackability in Code RL Training Environments - accessed 2026-07-30
- Benchmarking LLM-as-a-Judge for Long-Form Output Evaluation (LongJudgeBench) - accessed 2026-07-30
- Beyond the Leaderboard: A Synthesis of Tool-Use, Planning, and Reasoning Failures in LLM Agents (abstract) - accessed 2026-07-30

- Can LLM-as-a-Judge Reliably Verify Rubrics in Agentic Scenarios? (RuVerBench) - accessed 2026-07-30
- Can LLM-as-a-Judge Reliably Verify Rubrics in Agentic Scenarios? (RuVerBench, abstract) - accessed 2026-07-30
- CHERRL full text v1 - accessed 2026-07-30
- Claude Fable — Anthropic (official product page) - accessed 2026-07-30
- CodeJudgeBench: Benchmarking LLM-as-a-Judge for Coding Tasks - accessed 2026-07-30
- DeepSWE: Measuring Frontier Coding Agents on Original, Long-Horizon Engineering Tasks - accessed 2026-07-30
- Delayed Verification Destabilizes Multi-Agent LLM Belief (abstract) - accessed 2026-07-30
- Do You Need a Frontier Model as a Citation Verifier? Benchmarking Rubric LLMs for Deep-Research Source Attribution (abstract) - accessed 2026-07-30
- GitLab Advisory Database — CVE-2026-33068 (@anthropic-ai/claude-code) - accessed 2026-07-30
- GPT-5.6 — OpenAI (official index page; returned HTTP 403 to fetcher, exists) - accessed 2026-07-30
- Hallucination Cascade: Analyzing Error Propagation in Multi-Agent LLM Systems (abstract) - accessed 2026-07-30
- https://arxiv.org/abs/2507.15219 - accessed 2026-07-30
- https://arxiv.org/abs/2510.09023 - accessed 2026-07-30
- https://arxiv.org/abs/2606.26479 - accessed 2026-07-30
- Introducing Claude Opus 5 — Anthropic (official) - accessed 2026-07-30
- Introducing Claude Tag — Anthropic (official) - accessed 2026-07-30
- Introducing GPT-5.6 series: Sol, Terra and Luna — OpenAI Developer Community (Announcements) - accessed 2026-07-30
- Introducing the OpenHands Index (OpenHands blog) - accessed 2026-07-30
- Long-running Claude for scientific computing — Anthropic research (Mar 23, 2026) - accessed 2026-07-30
- Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity (METR) - accessed 2026-07-30
- More Convincing, Not More Correct (full text v1) - accessed 2026-07-30
- More Convincing, Not More Correct: Self-Play Reward Hacking of Reference-Free LLM Judges (abstract) - accessed 2026-07-30
- NIST SP 800-218A publication landing page - accessed 2026-07-30
- OWASP MCP Security Cheat Sheet - accessed 2026-07-30
- Position: Coding Benchmarks Are Misaligned with Agentic Software Engineering (abstract) - accessed 2026-07-30
- PromptArmor — Hijacking Claude Code via Injected Marketplace Plugins - accessed 2026-07-30
- Redeploying Claude Fable 5 — Anthropic (export-control timeline) - accessed 2026-07-30

- Reliability without Validity: A Systematic, Large-Scale Evaluation of LLM-as-a-Judge Models (abstract) - accessed 2026-07-30
- Reproducing, Analyzing, and Detecting Reward Hacking in Rubric-Based RL (CHERRL, abstract) - accessed 2026-07-30
- Reward Hacking Benchmark (full text v1) - accessed 2026-07-30
- Reward Hacking Benchmark: Measuring Exploits in LLM Agents with Tool Use (abstract) - accessed 2026-07-30
- SpecBench: Measuring Reward Hacking in Long-Horizon Coding Agents (abstract) - accessed 2026-07-30
- SWE-bench leaderboards (official site) - accessed 2026-07-30
- SWE-bench Pro Public Leaderboard (Scale) - accessed 2026-07-30
- The Verification Horizon: No Silver Bullet for Coding Agent Rewards (full text v2, Table 3) - accessed 2026-07-30
- Towards a Science of Scaling Agent Systems (Google Research blog) - accessed 2026-07-30
- We are Changing our Developer Productivity Experiment Design (METR) - accessed 2026-07-30

**Tier 3** (5)

- AI Engineer World's Fair 2026 — official event site - accessed 2026-07-30
- Claude v1 discovery: www.sophos.com/en-us/blog/inside-the-lethal-trifecta-blast-radius-reduction-in-ai-agent-deployments - accessed 2026-07-30
- CodeScaleBench (Sourcegraph blog) - accessed 2026-07-30
- https://github.com/openai/codex/issues/8759 - accessed 2026-07-30
- Olaf Alders — Claude Code: Anatomy of a Misfeature - accessed 2026-07-30

**Tier 4** (10)

- 5 Trends That Defined AI Engineering at World's Fair 2026 — Latent Space - accessed 2026-07-30
- AI Engineer conference talks - accessed 2026-07-30
- AIEWF Daily Dispatch: the great loops debate — Latent Space - accessed 2026-07-30
- Anthropic official YouTube channel - accessed 2026-07-30
- Codex from 0 to 10M Users: Building ChatGPT Work — Akshay Nathan (Latent Space) - accessed 2026-07-30
- Harness Engineering is not Enough — Dex Horthy, HumanLayer (AI Engineer YouTube) - accessed 2026-07-30
- Latent Space podcast - accessed 2026-07-30
- OpenAI official YouTube channel - accessed 2026-07-30
- OpenAI unveils ChatGPT Work agent, GPT-5.6 models now available — 9to5Mac - accessed 2026-07-30
- The new GPT-5.6 family: Luna, Terra, Sol — Simon Willison - accessed 2026-07-30

**Tier 5** (2)

- Software Factories, Light and Dark — Addy Osmani (talk write-up) - accessed 2026-07-30
- What to Watch from AIEWF 2026 (talk index) — Lawrence Wu - accessed 2026-07-30